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Bundle prices versus bundle quantities in uncertain fisheries

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Abstract

Many fisheries cannot be regulated one species at a time, and some fisheries manage multiple stocks under a single economic instrument. I study when such bundling is desirable and whether it should take the form of a shared quota or a shared landing fee (tax), extending the constant-escapement framework of Reed (1979) and Weitzman (2002) to a multispecies fishery under ecological uncertainty.

The central condition for a well-designed basket is equalizing marginal profits across species, a condition that combines biological productivity, harvesting costs, and prices. Under this criterion, both instruments are first-best when shocks are deterministic, but under uncertainty both become second-best, with the tax outperforming the quota. The main reason for this difference is that the quota must be set before shocks are realized. As the basket grows, the quota's disadvantage is diversified away, but a second cost, the mismatch from pooling heterogeneous species, grows with basket size. Larger baskets are not always better.

Bundling improves on species-specific management only when the monitoring costs it saves exceed this pooling loss.

1 Introduction

Allocating property rights through species-specific quotas has been shown to solve the common-pool resource problem in fisheries (Costello et al., 2016; FAO, 2025; Hilborn et al., 2020; Hoshino et al., 2020; Mangin et al., 2018). Without quotas, competing fishermen have the incentive to guarantee their revenue by harvesting more today, depleting the available biomass tomorrow (Clark, 1990; Gordon, 1954). Yet most commercial fisheries do not harvest one stock at a time (Spence et al., 2021), and only a small fraction operates with negligible incidental catch (Fulton et al., 2022). Accepting this reality, some fisheries regulate multiple fish stocks with a single economic instrument for the whole group (quota basket) (Sanchirico et al., 2006). In contrast, quota baskets have historically been dismissed as a solution because they risk overfishing vulnerable species (Bonzon et al., 2013; Clark and Kirkwood, 1986). Then, how can we reconcile the evidence that quota baskets can effectively conserve multiple species while maintaining economic viability (Collado et al., 2021; Karr et al., 2021; Kritzer et al., 2023)? What are the key conditions for a successful quota basket?

Using species-specific regulation for multispecies fisheries poses inherent challenges because of joint production, limited selectivity, imperfect information about harvest composition, and fishermen's behavior (Branch, 2009; Christensen and Pauly, 2004; Copes, 1986; May and Clark, 1979; Nakamura, 2015; Pauly et al., 1998; Sanchirico et al., 2006). First, it ignores natural clusters (shoals) and species interactions through ecology, regulation, and non-selective fishing gear. A consequence is omitting the trade-off between sustainability and profit maximization caused by the choke species: the most restrictive species lead to premature season closure (Abbott and Wilen, 2009) and a shortage of quota that blocks the extraction of non-vulnerable species (Hatcher, 2022; Ulrich et al., 2017; Wo et al., 2022). This is the case of Canary and Yelloweye rockfish in California: the protection of the first forgoes revenue margins of the second, and a higher catch of the second endangers the rebuilding of the first.

Second, each species caught jointly is itself a common-pool resource, a separate market failure. Tinbergen's (1952) rule would call for a corrective instrument for each failure, but managers are constrained by how many economic instruments they can credibly design, monitor, enforce, and govern. A fully-implemented regulation for just one species is data-intensive (Mangin et al., 2018; Su et al., 2020), costly to administer (Karr et al., 2021), and prone to political instability (OECD, 2003). Even if the resources are available, species are hard to identify at port (Pacific Fishery Management Council, 2024; Squires et al., 1998) or aggregated in the datasets (Su et al., 2020; Sun et al., 2023). Such problems are especially pronounced in Chinese fisheries, which often harvest a broad mix of species with minimal selectivity, leading to indiscriminate removal across trophic levels (X. Jin et al., 2020; Kritzer et al., 2023; Szuwalski et al., 2017). Low selectivity hampers control and monitoring efforts (Sun et al., 2023; Ye et al., 2013), while aggregation in official landings data complicates identifying stock/harvest composition (Cao et al., 2017; Pauly and Froese, 2012). These issues limit the feasibility of traditional species-specific management that relies on precise stock assessments.

Third, using specific instruments for a group is not limited to task difficulty; it also extends to concrete harm resulting from the mismatch: it induces discarding of the low-value species[1] (Copes, 1986; Ulrich et al., 2017), effort reallocation toward unregulated stocks (Smith et al., 2009), and intertemporal arbitrage (Holzer and DePiper, 2019). The mismatch between the number of instruments and species degrades trophic relationships and ecosystem integrity, erodes the welfare of the whole sector, and gains efficiency for one species at the cost of the remaining stocks.

[1] From now on, high-grading

This aggregation problem belongs to a broader class of second-best regulation problems in which a single instrument must address multiple market failures. In environmental economics, this question has been studied for uniform versus differentiated taxes in a static setting across heterogeneous pollution sources (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986), and for the choice between prices, quantities, and mixed policies when multiple pollutants are jointly regulated (Ambec and Coria, 2013; Hoel and Karp, 2002). Fisheries is not left behind, as many approaches have been proposed to deal with multi-species fisheries, but I want to highlight a few. First, Singh and Weninger (2024) studied cross-species flexibility in individual quotas: a fraction of one species' quota can be applied to another. Their framework retains species-specific quotas and asks how much flexibility to allow. More closely related, Hatcher (2022) proposes managing by setting quota-lease prices to implicitly equalize marginal profits across species in a portfolio. The diversification result also connects to portfolio approaches to fisheries management (Edwards et al., 2004; D. Jin et al., 2016; Sanchirico et al., 2008), which exploit covariance across stocks to manage risk. All these methods are attractive but prohibitive in terms of scientific capability, information, and cost for many regulators globally.

Unlike these solutions, quota baskets do not rely on rigid single-species frameworks or rich datasets (Collado et al., 2024). They consider full bundling of the species under a single instrument, allowing flexibility in harvest allocation by trading stringency for convenience. However, the theoretical and empirical research on this subject is scarce. Clark's earliest theory (1990) showed that joint harvesting could not sustain both species in the short run, depleting the more vulnerable stock in a static open-access setting. Extending the analysis dynamically, Clark demonstrated that joint harvesting may be feasible but leads to an unstable equilibrium. In a non-fishery setting, Weitzman (1998) examined biodiversity preservation under budget constraints and showed that optimal bundling strategies may imply sacrificing some species to save others. Recently, Collado et al. (2021) analyzed fisher behavior under multiple basket allocations and varying fishing gear choice in a modified Gordon–Schaefer model. Their results suggest that well-designed baskets can maintain moderate overall stock survival, whereas poorly designed configurations risk population collapse of the vulnerable species or of all species in the group.

Finally, consensus remains lacking on how to group species or the conditions needed to achieve a welfare-maximizing quota basket, except in some empirical studies. For the first case, previous research recommends grouping species by similarity in ecological and economic traits (Collado et al., 2021; Costello and Szuwalski, 2018), similar vulnerability indicators (catchability over growth rate)(Collado et al., 2024), or relying on the expertise of local fishermen (EDF, 2020). Using empirical data from Belizean fisheries (United Nations Conference on Trade and Development, 2022), Collado et al. (2024) determined that the optimal quota basket was set at approximately 30% of the sum of individual maximum sustainable yields. These efforts build intuition but do not provide a definitive answer, especially in more complex settings such as managing ecological uncertainty.

This paper continues the search for second-best instruments by building the quota basket theory and a tax basket, which I introduce, that applies a single landing fee to a group of species. Both bundle multiple common-pool market failures under a shared instrument. The question is, under what conditions do we need to establish a welfare-maximizing bundle instrument? If lumping works, how does the choice between price and quantity matter under ecological uncertainty?

Building on Reed's (1979) and Weitzman's (2002) constant-escapement framework and information structure, I extend their analysis to a multispecies fishery for one regulator, one fisherman, and one basket. These models are a propitious stage because (i) fisheries are dynamic, (ii) the optimal escapement is a "fixed" and widely accepted policy outcome in fisheries, and (iii) the information asymmetry between the agent and regulator simplifies the uncertainty analysis. The model does not incorporate management costs, fishing gear, biological interactions across bundled species, or strategic interactions among fishermen to isolate the instrument's role under ecological uncertainty. For simplicity, I also assume that all species in the basket have commercial value for the fisherman (in other words, they belong to the fishable set). In consequence, I don't cover topics such as bycatch and discarding.

My main result is that multispecies fisheries have one fundamental condition: equalize the marginal profit of the last ton caught for each species. As in pollutant markets, if all species are desirable, the fisherman maximizes total profits by applying an equimarginal principle. While different economic and ecological traits matter, marginal profits combine them. In an open-access setting, a myopic fisherman maximizes current profits by setting marginal profits equal to marginal costs. Under stock-dependent costs, the remaining or unfished biomass (escapement) of the fisherman's optimum may not coincide with sustainable escapement. A tax or quota forces the internalization of these intertemporal costs. Given my assumptions, I expand this framework to multiple marginal profits and escapements. Through this condition, I make five contributions:

First, in a deterministic setting, I show that equating marginal profits across species at their respective optimal escapement levels implements the first-best outcome for both instruments. Second, under ecological uncertainty, the tax baskets remain first-best if the marginal profits remain identical, while quota baskets become second-best. This extends Weitzman's (2002) single-species landing-fee result to a multispecies setting. Third, when species within a basket differ in marginal profits, both instruments are second-best, but tax baskets continue to outperform quota baskets. In this case, the optimal uniform instrument corresponds to a weighted average of species-specific marginal profits, mirroring the optimal uniform tax across heterogeneous pollutants (Ambec and Coria, 2013; Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). Fourth, the performance difference between quota and tax baskets shrinks as the basket size increases. This diversification result connects the basket framework to portfolio theory (Edwards et al., 2004; D. Jin et al., 2016; Markowitz, 1952; Sanchirico et al., 2008). The novelty is reducing variance by adding species to the group rather than optimizing species-specific weights. Fifth, I decompose the basket's regulatory loss, relative to species-specific first-best instruments, into a pooling component (the mismatch from applying one instrument to heterogeneous species), which grows with the number of species, and an uncertainty component (the timing distortion) which is diversified away as the basket grows. The two move in opposite directions, so bundling is welfare-improving relative to species-specific management only when the monitoring and administration cost savings outweigh the pooling loss.

The paper also speaks to the literature on partial flexibility in multispecies quota systems. Where Singh and Weninger characterize the optimal partial flexibility parameter conditional on a species-specific IFQ design, I derive the conditions for full bundling and compare the resulting quota basket with its tax counterpart. A complementary venue is Hatcher (2022), who models quota prices in a multispecies fishery with choke species and discarding. Like him, I use marginal quota values that adjust with the overall profitability of the catch portfolio.

Finally, the paper contributes to the fisheries-under-uncertainty literature (Clark and Kirkwood, 1986; Hannesson and Kennedy, 2003; Reed, 1979; Sethi et al., 2005). The analysis shows that, even under uncertainty, basket instruments can be designed using a well-defined optimality criterion rather than ad hoc grouping rules.

The remainder of the paper is organized as follows. Section 2 presents the model and theoretical results. Section 3 reports the numerical results. Section 4 discusses policy implications and concludes.

2 Model

Throughout the paper, $t = 0, 1, \ldots$ denotes time, and $i = 1, 2, \ldots, N$ indexes the different species. The model assumptions rely on two cornerstones: (i) the intertemporal profits for a particular species are maximized with fixed escapement over time in deterministic and stochastic settings (Reed, 1979), and (ii) a fixed landing fee induces the fisherman to stop fishing at the optimal escapement [2] (Weitzman, 2002). In this context, my assumptions are the following:

i. The average stock recruitment function $G^i(y^i_{t-1})$ is differentiable, concave $\frac{\partial^2 G(\psi^i, y^i_{t-1})}{\partial (y^i_{t-1})^2} \leq 0$, increasing $\frac{\partial G(\psi^i, y^i_{t-1})}{\partial y^i_{t-1}} > 0$ and bounded. It depends on the remaining stock or escapement y^i_{t-1} of the previous period.

ii. The marginal harvesting cost is decreasing in available stock s^i: $C(s^i) > 0$, $C'(s^i) < 0$,

iii. The random ecological $\{Z^i_t\}$ are independent and identically distributed over time on some interval $[a, b]$ [3], with $\mathbb{E}[Z^i_t] = 1$ and $\mathbb{E}|\ln Z^i_t| < \infty$,

[2] An alternative proof can be found in Appendix A.

[3] The population is self-sustaining within the intervals $m = \max\{bG(s) : bG(s) \geq s, s \geq 0\}$ and $r = \max\{aG(s) : aG(s) \geq s, s \geq 0\}$

iv. The prices are strictly positive for all species in the basket, so discards are not considered, as all harvest is valuable,

v. Prices p^i, cost parameters θ^i, growth functions G^i, the discount factor β, and the shock distribution are time-invariant,

vi. For simplicity, the fisherman chooses the optimal fishing technology, allowing differentiation of the cost coefficient across species within a basket[4],

vii. Weak exogeneity of stock on ecological shock and choice variables: the shock (Z_t^i) has zero conditional expectation given the present and past values of X_t^i and the choice variables, and

viii. Species' life cycles are independent, so the growth function does not depend on the escapement of other stocks.

ix. All species belong to a fishable set $\mathcal{F}$ (later defined in Lemma 2.1) to guarantee that all species remain in the basket.

I also adopt Weitzman's (2002) information framework, which combines a static optimization problem for a myopic fisherman with a dynamic stochastic optimization problem for the social planner:

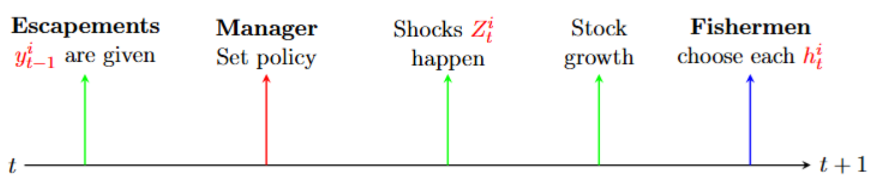

Figure 1: Information flow for each species i between periods t and $t+1$

At each period t, the following sequence of events occurs:

1. Escapements y_{t-1}^i for each species i are predetermined from the choices on the previous period.

2. The social planner sets a basket policy (a quota ω_t or a tax τ_t), unaware of stock composition.

3. The ecological shocks Z_t^i are realized. Each species grows.

[4]Collado et al. 2024 proved that the fisherman chooses the harvest technology that maximizes basket profits.

4. The fisherman observes the realized stock and chooses the harvest h_t^i, given the basket policy set by the planner.

The key friction is that the planner chooses a basket policy to address multiple common-pool resource problems before shocks are realized, whereas harvest decisions are made after shocks are realized. Given the economic instrument, a myopic fisherman observes the realized shocks and chooses harvest to maximize current profit:

1. Under a tax basket:

$$\max_{\{h_t^i\}_{i=1}^N} \int_{X_t^i - h_t^i}^{X_t^i} (p^i - \tau_t - C(s^i, \theta^i)) ds^i \tag{1}$$

2. Under a quota basket:

$$\begin{aligned} &\max_{\{h_t^i\}_{i=1}^N} \sum_{i=1}^N \int_{X_t^i - h_t^i}^{X_t^i} (p^i - C(s^i, \theta^i)) ds^i \\ &\text{subject to} \quad \omega = \sum_{i=1}^N h_t^i \end{aligned} \tag{2}$$

where X_t^i is the stock at the start of period t, h_t^i is the harvest, p^i is the price, and (θ^i, ψ^i) are cost and biological parameters, which are fixed over time. Due to the importance of the escapement y^i as a reference for species-specific profit maximization, I re-express the integral bounds using the escapement definition $y_t^i = X_t^i - h_t^i$:

$$\sum_{i=1}^N \int_{X_t^i - h_t^i}^{X_t^i} \pi^i(s^i) ds^i \rightarrow \sum_{i=1}^N \int_{y_t^i}^{X_t^i} \pi^i(s^i) ds^i$$

On the other hand, the forward-looking social planner sets the instrument before (ω_t or τ_t) shocks are realized to maximize discounted profits, taking the fisherman's decision rule for each species i as given:

1. Under a tax basket:

$$E\left[\sum_{t=0}^{\infty} \beta^t \sum_{i=0}^{N} \underbrace{\int_{X_t^i - h_t^i(\tau)}^{X_t^i} (p^i - \tau_t - C^i(s^i)) ds^i}_{\text{Species i profit}} + \underbrace{\int_{X_t^i - h_t^i(\tau)}^{X_t^i} \tau_t ds^i(\tau)}_{\text{Species i tax revenue}}\right] \tag{3}$$

Notice the tax terms in the profit and revenue components cancel.

2. Under a quota basket:

$$
\begin{gathered}
E\left[\sum_{t=0}^{\infty}\beta^t\sum_{i=0}^{N}\int_{X_t^i-h_t^i(\omega)}^{X_t^i}(p^i-C^i(s^i,\theta^i))ds^i\right] \\
\text{subject to} \quad X_t^i=Z_t^iG^i(y_{t-1}^i,\psi^i), \quad \omega=\sum_{i=1}^{N}h_t^i
\end{gathered} \tag{4}
$$

where $\beta \in (0,1)$ is the discount factor and ψ^i are the biological parameters

To reduce notational clutter, I omit explicit references to biological (ψ^i) and cost (θ^i) parameters unless necessary. Thus, I express the cost function as $C^i(s^i)$ and growth as $G^i(y_{t-1}^i)$ to denote different parameters across species.

2.1 Welfare analysis

Assuming a risk-neutral government, the objective of this paper is the welfare for each instrument:

$$\boxed{\Delta=\mathcal{W}^{\tau}-\mathcal{W}^{\omega}} \tag{5}$$

To calculate this difference, I need to find

1. The fisherman's static decision rule y^i under each instrument.
2. The planner's optimal instrument (tax or quota) incorporating (1).
3. Calculate the fisherman's escapement with the optimal instrument found in (2).
4. Calculate the welfare W under each instrument.

Throughout the propositions, a first-best outcome refers to a policy that maximizes intertemporal profits under the relevant information structure in the deterministic setting. A second-best outcome refers to a policy that maximizes intertemporal profits subject to informational constraints, but yields strictly lower expected profits than the deterministic benchmark.

2.1.1 Fisherman's decision rule

From the FOC of Equation 3 concerning each harvest, I obtain the optimality condition for the current period:

$$p^1-C^1\left(X_t^1-h_t^1\right)=p^2-C^2\left(X_t^2-h_t^2\right)=\cdots=p^n-C^n\left(X_t^n-h_t^n\right)=\tau_t$$

Using the definition of escapement $y_t^i=X_t^1-h_t^1$, I can re-express this condition as:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = \tau_t \tag{6}$$

Notice that Equation 6 is the equimarginal principle: the total profits are maximized when the marginal profit of the last harvested unit for each species i is equal. The fisher sets the escapement decision:

$$\begin{aligned} p^i - \tau_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - \tau_t \\ y_t^i &= C^{i-1}(p^i - \tau_t) \end{aligned} \tag{7}$$

For the static problem, notice that tax does not depend on the state variables (and I verify later in the planner's problem). In Equation 4, the fisherman faces the same maximization problem but for a given quota basket ω_t. The optimality condition in this case is:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = \lambda_t \tag{8}$$

Obtaining the same fisherman's escapement decision $y_t^i = C^{i-1}(p^i - \lambda_t)$. Notice, however, that the shadow value is a function of the vector $\vec{X}$ and the quota basket ω_t because

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i y_t^i \\ \omega_t &= \sum_i X_t^i - \sum_i C^{i-1}(p^i - \lambda_t) \\ \omega_t &= \sum_i X_t^i - \sum_i y^i(\lambda_t) \\ \sum_i y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \sum_i y^i(\lambda_t) &= RHS_t \\ \lambda_t &= \lambda(\vec{X}_t, \omega_t) \end{aligned} \tag{9}$$

The fisherman decides their total harvest by equalizing the marginal profits $m\pi_t = \tau_t = \lambda_t(\vec{X}_t, \omega_t)$ across all species within the basket. Then, the fisherman's decision under a quota basket is:

$$y_t^i = C^{i-1}(p^i - \lambda(\vec{X}_t, \omega_t)) \tag{10}$$

In this case, the shadow value depends on the state of all stocks, increasing its complexity relative to a tax basket. In any case, the key instrument to influence the optimality decision is the marginal profits, and tax and quota are vehicles to set them:

Lemma 2.1 (Fisherman's optimality condition) *The optimality condition for harvesting N species in the fishable set $\mathcal{F} = \{i : p^i - m\pi > C^i(K^i)\}$* [5] *is:*

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = m\pi_t$$

Through the stock-dependent cost function, the optimality condition incorporates all economic and biological parameters of all species.

The equimarginal principle holds: the total profits are maximized when the marginal profit $m\pi_t$ of the last harvested unit for each species i is equal.

2.1.2 Optimal marginal profits

The social planner incorporates the above fisherman's escapement decision into his dynamic optimization problem:

$$\begin{aligned} V_t(\vec{y_t}) = \max_{m\pi_t \geq 0}\ & \mathbb{E}\left\{\sum_{i=1}^{N} \int_{y_t^i(m\pi_t)}^{Z_t^i G^i(y_{t-1}^i(m\pi_{t-1}))} \left(p^i - C^i(s^i)\right) ds^i + \beta V_{t+1}(\vec{X_t})\right\} \\ \text{s.t. } & X_t^i = Z_t^i G^i(y_{t-1}^i) \end{aligned} \tag{11}$$

I take the FOC with respect to $m\pi_t$, and obtain the optimal marginal profit:

Proposition 1 (Welfare-maximizing $m\pi$) *Consider N heterogeneous, independent stocks with cost function C^i, i.i.d. ecological shocks Z_t^i with $\mathbb{E}[Z_t^i] = 1$, and growth functions $G^i(y_{t-1}^i)$. Assume the interior regularity condition*

$$\max_i m\pi^{i*} \ < \ m\pi_{\max}^F \ := \ \min_i \bigl(p^i - C^i(K^i)\bigr)$$

so that all species remain in the fishable set on $[\min_i m\pi^{i*}, \max_i m\pi^{i*}]$. Then, the optimal $m\pi_t^+$ is determined by the following implicit function:

$$m\pi_t^+ = \beta \frac{1}{\sum_{j=1}^{N} y_{m\pi_t^+}^j} \mathbb{E}\Big[\sum_{i=1}^{N} (p^i - C^i(Z_t^i G^i(y_t^i))) Z_t^i G_{y_t^i}^i y_{m\pi_t^+}^i\Big] \tag{12}$$

Let's work the Euler equation from Appendix B a little further:

[5] where K^i is the carrying capacity of species i

$$
\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y_t^i))y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^iG^i(y_t^i)))Z_t^iG^i_{y_t^i}y^i_{m\pi_t}\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(C^{i-1}(p^i - m\pi_t)))y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^iG^i(y_t^i)))Z_t^iG^i_{y_t^i}y^i_{m\pi_t}\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - (p^i - m\pi_t))y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^iG^i(y_t^i)))Z_t^iG^i_{y_t^i}y^i_{m\pi_t}\Big] \\
\sum_{i=1}^{N} m\pi_t y^i_{m\pi_t} &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^iG^i(y_t^i)))Z_t^iG^i_{y_t^i}y^i_{m\pi_t}\Big] \\
m\pi_t \sum_{i=1}^{N} y^i_{m\pi_t} &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^iG^i(y_t^i)))Z_t^iG^i_{y_t^i}y^i_{m\pi_t}\Big] \\
m\pi_t^+ &= \beta\frac{1}{\sum_{j=1}^{N} y^j_{m\pi_t}}\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^iG^i(y_t^i)))Z_t^iG^i_{y_t^i}y^i_{m\pi_t}\Big]
\end{aligned}
\tag{13}
$$

The Euler equation balances current marginal profit with the weighted average of next-period marginal profits.

Corollary 1.1 (The optimal $m\pi_t$ is constant) *The optimal marginal profit is constant over time: $m\pi_t = m\pi_{t+1} = m\pi$.*

From expression 13, it is easy to observe that $m\pi \perp \vec{X}_t$. Because all species remain in the fishable set, the fisherman also equalizes marginal profits in period $t+1$:

$$
\begin{aligned}
m\pi_t &= \sum_{i=1}^{N}\mathbb{E}\Big[m\pi_{t+1}\Big]\frac{y^i_{m\pi_t}}{\sum_{j=1}^{N} y^j_{m\pi_t}} \\
m\pi_t &= \sum_{i=1}^{N}\mathbb{E}\Big[m\pi_{t+1}\Big]\rho_t^i \\
m\pi_t &= \mathbb{E}\Big[m\pi_{t+1}\Big]\sum_{i=1}^{N}\rho_t^i \\
m\pi_t^+ &= \mathbb{E}\Big[m\pi_{t+1}\Big] \qquad [\text{because } \sum_i \rho_t^i = 1]
\end{aligned}
$$

Because $X_t^i > y_t^i$ for $\forall i, t$ (according to assumption iii), the summation of weights ρ_t^i is 1 because I assume all species remain in the fishable set. Then the Euler equation equates current marginal profits with next-period expected marginal profits. On average, $m\pi_t$ is fixed over time. From this expression, it is possible to obtain the following remarks:

Remark 1.1 (Deterministic case) *If $Z_y^i \equiv 1$, then $m\pi_t = m\pi_{t+1} = m\pi$.*

If the cost function follows assumption ii, I can recover a marginal profit unaffected by the ecological perturbations,

Corollary 1.2 (Specific cost function recovers the deterministic case) *If the cost function is decreasing with respect to X^i, then $m\pi_t = m\pi_{t+1} = m\pi$.*

Let's replace $C^i(s_t^i) = \frac{\theta^i}{s_t^i}$ in Proposition 1:

$$\begin{aligned}
m\pi_t^+ &= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i - C^i(Z_t^i G^i(y_t^i))) Z_t^i G_{y_t^i}^i y_{m\pi_t}^i\Big] \\
&= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i - \frac{\theta^i}{Z_t^i G^i(y_t^i)}) Z_t^i G_{y_t^i}^i y_{m\pi_t}^i\Big] \\
&= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i Z_t^i - \frac{\theta^i Z_t^i}{Z_t^i G^i(y_t^i)}) G_{y_t^i}^i y_{m\pi_t}^i\Big] \\
&= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \sum_{i=1}^N (p^i - \frac{\theta^i}{G^i(y_t^i)}) G_{y_t^i}^i y_{m\pi_t}^i \\
&= m\pi_{t+1} \sum_{i=1}^N \rho_t^i \\
&= m\pi_{t+1}
\end{aligned}$$

Under this cost function, the marginal profit behaves as in the deterministic case. If the cost function is $C^i(s_t^i) = \frac{\theta^i}{(s_t^i)^\alpha}$, then the marginal profit is a lower bound (see Appendix B).

Returning to our static fisherman's problem: for now, the planner knows the fisherman's escapement decision differs by economic instrument. In summary, the tax instrument does not seem to depend on the states, while the quota includes the states through a stochastic shadow value. In consequence, the fisherman's escapement is deterministic under a tax ($\vec{\bar{y}}$) and stochastic under a quota y. However, in a deterministic case, the shadow value is also deterministic because $\lambda(\vec{X}_t, \omega_t) = \lambda(Z_t^i \vec{G^i}(.)_t, \omega_t) = \lambda(1\ \vec{G^i}(.)_t$. Then, I can conjecture that the shadow value and tax basket behave the same way: $\tau_t = \lambda(\omega_t)$. In addition, if all species have the same marginal profit under their respective optimal escapements, then both instruments are first-best to target the optimal $m\pi^+$:

Corollary 1.3 (First-best basket instruments) *Given the core assumptions, the tax basket or a quota basket implements a first-best $m\pi^+$ if and only if $p^i - C^i(y^{i*}) = p^j - C^j(y^{j*})$ for all $i \neq j$ in a deterministic setting.*

Note that if species have identical marginal profits at their respective optimal escapement levels, I can apply Reed's maximization $\frac{p^i - C^i(Z^i G^i(y^{i*}))}{p^i - C(y^{i*})} G'(y^{i*}) = \frac{1}{\beta}$ condition to each species to maximize profits over time:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = \tau_t = \lambda(\omega_t)$$
$$p^1 - C^1\left(y_t^{1*}\right) = p^2 - C^2\left(y_t^{2*}\right) = \cdots = p^n - C^n\left(y_t^{n*}\right) = \tau^* \lambda(\omega_t)$$
$$\beta(p^1 - C^1(G^1(y^{1*}))G'(y^{1*}) = \cdots = \beta(p^n - C^n(G^n(y^{n*}))G'(y^{n*}) = \tau^* = \lambda(\omega_t)$$

Even if species' profits are heterogeneous, a fixed tax or shadow value achieves a constant optimal escapement or Reed's first best. However, this analysis is incomplete because I need to show how the planner chooses the instrument in the dynamic setting, whether choosing the optimal shadow value is equivalent to choosing the optimal ω_t^+ to achieve the optimal $m\pi^+$, and the duality between τ_t^+ and λ_t^+. Before jumping to the next section, the following corollary would prove useful:

Corollary 1.4 (Next-period weights) *It is possible to re-express the next-period total profit weights as a function of the escapement and cost parameter if the cost function is* $C^i(s_t^i) = \frac{\theta^i}{s_t^i}$*:*

$$m\pi^+ = \frac{\sum_{i=1}^{N} \pi^i\big(Y^i(m\pi^+)\big) \frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}}{\sum_{i=1}^{N} \frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}} \tag{14}$$

If the cost function is $C^i(s_t^i) = \frac{\theta^i}{s_t^i}$, then $y^i_{m\pi_t} = \frac{\theta^i}{(p^i - m\pi)^2}$

$$\begin{aligned} m\pi_t^+ &= \sum_{i=1}^{N} m\pi_{t+1} \frac{y^i_{m\pi_t}}{\sum_{j=1}^{N} y^j_{m\pi_t}} \\ &= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\theta^i}{(p^i - m\pi)^2}}{\sum_{j=1}^{N} \frac{\theta^i}{(p^i - m\pi)^2}} \end{aligned}$$

I know the fisherman decision is $p^i - m\pi = C^i(y^i)$ and $C^i(y^i) = \frac{\theta^i}{y^i}$:

$$
\begin{aligned}
m\pi_t^+ &= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\theta^i}{(p^i - m\pi)^2}}{\sum_{j=1}^{N} \frac{\theta^i}{(p^i - m\pi)^2}} \\
&= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\theta^i}{C^i(y^i)^2}}{\sum_{j=1}^{N} \frac{\theta^i}{C^i(y^i)^2}} \\
&= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}}{\sum_{j=1}^{N} \frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}}
\end{aligned}
$$

2.1.3 Planner's problem

After studying the fisherman's decision and recognizing how the dynamic selection $m\pi^+$ can influence the harvest decision, the planner chooses the magnitude of the tax or quota basket to maximize intertemporal profits. For the tax basket, the solution of the dynamic problem is straightforward because it follows the same procedure as in Appendix B, after replacing $m\pi_t = \tau_t$. Because the tax is independent of the state variable in both the fisherman's and planner's problems, it holds the same optimality condition for the welfare-maximizing marginal profit of equation 13 in Proposition 1

Proposition 2 (Welfare-maximizing tax basket) *Given the assumptions for N heterogeneous and independent stocks, species-specific optimal taxes τ^{i*} and interior regularity condition*

$$
\max_i \tau^{i*} \; < \; \tau_{\max}^F \; := \; \min_i \big(p^i - C^i(K^i)\big),
$$

[6]*, the welfare-maximizing tax basket τ^+ is characterized by the same implicit equation as in Proposition 1:*

$$
\tau_t^+ = \beta \frac{1}{\sum_{j=1}^{N} y_{\tau_t^+}^j} \mathbb{E}\Big[\sum_{i=1}^{N} (p^i - C^i(Z_t^i G^i(y_t^i))) Z_t^i G_{y_t^i}^i y_{\tau_t^+}^i\Big] \tag{15}
$$

Welfare is strictly lower than the summation of species-specific first-best whenever the τ^{i*} are not all equal. Then, τ^+ is the second-best. The tax basket τ^+ also satisfies corollaries 1.1 to 1.4.

By replacing $m\pi_t^+ = m\tau_t^+$ in Proposition 1 and all its related corollaries it is easy to observe that the optimal tax basket mirrors the optimal marginal profit Euler equation, is fixed over time (especially if the cost function is $C^i = \theta^i/s^i$, is a first-best instrument when all species

[6]So that all species remain in the fishable set on $[\min_i \tau^{i*}, \max_i \tau^{i*}]$

have the same marginal profit under optimal escapement, and its weight depend on the escapement and the cost parameter (for a complete derivation, see Appendix B.3). By inspection,

Remark 2.1 (Suboptimal tax basket) *The suboptimal tax basket implements a suboptimal constant escapement for all species.*

More interestingly, the selection of the optimal tax remains fixed even in a stochastic setting:

Corollary 2.1 *Given the same assumptions and conditions of Corollary 1.3, the tax-basket also implements the first-best constant $m\pi^+$ in a stochastic setting.*

This result is not surprising, given that the tax doesn't depend on the state variables.

$$\begin{aligned} \int_{y_t^i}^{X_t^i} \left(p^i - \frac{\theta^i}{s}\right) ds &= p^i(X_t^i - y_t^i) - \theta^i \ln(X_t^i/y_t^i) \\ &= p^i X - \theta^i \ln X - p^i y + \theta^i \ln y \\ &= A^i(X_t^i) + \phi^i(y_t^i), \end{aligned} \tag{16}$$

after substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, using $\mathbb{E}[Z_t^i] = 1$ and $\mathbb{E}|\ln Z_t^i| < \infty$., summing across period and species is possible to obtain:

$$\mathcal{W} = \sum_i \underbrace{[A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \underbrace{[\phi^i(y_t^i) + \beta a^i(y_t^i)]}_{=:w^i(y_t^i)} \tag{17}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$ and $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$. While this expression is valid for a tax or quota, notice that substituting $y_t^i = Y^i(\tau_t)$, the escapement is constant over time (not optimal), and the term w^i is also constant, thanks to the boundaries on $G(.)$, $\beta < 1$, and the fishable set. In other words, for the tax basket, the same problem repeats each period, and it has the same solution.

Taking the derivative recovers the Euler equation from Proposition 2,

Corollary 2.2 (Successive myopic problems recover 15) *$\mathcal{W}$ is differentiable on the interior of the fishable range with*

$$\mathcal{W}'(\tau) = \sum_{i=1}^{N} \left[\pi_{t+1}^i\big(Y^i(\tau)\big) - \tau\right] y_{\tau_t}^i / \sum_i y_{\tau_t}^i, \tag{18}$$

so $\mathcal{W}'(\tau) = 0$ is exactly the implicit equation 15 defining τ^+.

Differentiating $w^i(y) = \phi^i(y) + \beta a^i(y)$,

$$\begin{aligned} w^{i\prime}(y) &= -\Big(p^i - \frac{\theta^i}{y}\Big) + \beta\Big(p^i - \frac{\theta^i}{G^i(y)}\Big)G^{i\prime}(y) \\ &= \pi^i_{t+1}(y) - \big(p^i - C^i(y)\big) \end{aligned}$$

Since $p^i - C^i(Y^i(\tau)) = \tau$, the chain rule gives 18 and Proposition 2. This corollary verifies that there are no kinks in the interior (see the full derivation in Appendix C.2. Corollary 2.2 is an input to prove the existence, uniqueness, and interiority of the tax basket [7]:

Corollary 2.3 (Existence, uniqueness and interiority of τ^+) *By Proposition 2, Corollary 2.2, regularity condition* $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$ *and cost function* $C^i = \theta^i/s^i$*, and second-order condition*

$$y\,\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) \;\le\; p^i + \tau \tag{19}$$

The optimal tax basket exists, is unique, and is interior.

The full proof can be found in Appendix D. Equation 19 is a second-order condition that guarantees the global concavity and complements the fishable-set assumption (first-order condition) to get a unique optimal tax basket. The fishable-set condition keeps every species harvested as τ varies, and 19 keeps each species' contribution concave. Both guarantee that a basket is not too heterogeneous, meaning there is some truth in the similarity condition across species within the bundle (Collado et al., 2021, 2024; Costello and Szuwalski, 2018; Kritzer et al., 2023).

Remark 2.2 (Information requirements) *The social planner does not require contemporaneous stock information to determine* τ^+ *or* $\bar{\tau}^*$*.*

Taken together, these results show that tax baskets preserve the constant-escapement structure under uncertainty but generally implement a second-best allocation when species differ in marginal profits. First-best outcomes arise only when marginal profits are identical across species at their respective optima. However, data collection is necessary if $X^i > y^i(\tau^+)$ is not profitable for each i, meaning the interiority condition is broken. Because species may enter and exit the fishable set, a dynamic tax basket is required:

Proposition 3 (Dynamic tax basket) *A tax basket is dynamic if the interiority condition (Appendix D.4) is broken.*

[7] The corner solutions and their implications are not part of this paper but will be discussed in a future publication.

For example, if $p^i - \theta^i/K^i < \tau^{j*}$ for some pair (i, j), then taxes near τ^{j*} push species i out of the fishable set; then W has a kink where the current payoff changes, and corner solutions are possible. In that case, the policymaker can implement a tax basket that changes over time according to the species that participate in the fishable set each period t.

Without loss of generality, assume all potential N species that can participate in the basket have different individual optimal taxes $\tau^{1*} \le \cdots \le \tau^{N*}$, and there is no regularity condition. Then, the optimality tax-basket condition in 15 is maintained, but it changes over time as the set of fishable species changes each period. Because the number of participant elements changes, the tax basket changes to guarantee profit maximization. However, in this case, the manager needs to invest in gathering information on participant species. In this scenario, bycatch reignites the tradeoff between conservation (designing the policy to include all species) and profit maximization (designing the policy to include only the potentially fishable species).

The quota basket is definitively dynamic. The essential difference with the tax basket is that a single quota constrains only aggregate harvest, so the split across species is the fisherman's choice; the shadow value λ_t is not a policy instrument but an equilibrium object that depends on the current stocks and the quota, $\lambda_t = \lambda(\mathbf{X}_t, \omega_t)$.

Proposition 4 (Welfare-maximizing quota basket) *Under the setup of Proposition 2, the welfare-maximizing quota ω^+ set before the shocks are realized satisfies*

$$\mathbb{E}\Big[\lambda_t - \beta \sum_i \rho_t^i V_{y^i}(\vec{y}_t)\Big] = 0, \qquad \rho_t^i := \frac{Y_\lambda^i(\lambda_t)}{\sum_j Y_\lambda^j(\lambda_t)} = \frac{(y_t^i)^2/\theta^i}{\sum_j (y_t^j)^2/\theta^j}, \tag{20}$$

where $\lambda_t = \lambda(\mathbf{X}_t, \omega_t)$ is the fisherman's shadow value, and V is the planner's next-period value function, and

$$V_{y_{t-1}^k}(\vec{y}_{t-1}) = \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \tag{21}$$

The derivation is available in Appendix E.3. Equations 20 and 21 are the quota-equivalent of the tax first-order condition 15. 20 equates, in expectation, the current shadow value of an extra unit of aggregate harvest with the ρ-weighted continuation value of the escapement. 21 is the shadow value of one extra fish in stock k minus the λ_t of the escapement it displaces, plus its continuation value. Separating the optimality condition into two parts is useful because 20 helps to visualize the role of the ecological shocks, and 21 reveals what happens when there is a deviation between the shadow value chosen by the manager, λ_t^+, and the effective shadow value the fisherman selects after the shocks, λ_t^F. Before discussing how the planner's capacity to know the ecological shocks causes a mismatch across shadow values, I need to establish that choosing λ^+ is equivalent to choosing τ^+:

To compare instruments and to characterize the quota's properties, I exploit the duality between the two problems.

Corollary 4.1 (Duality of λ^+ and τ^+) *Choosing the optimal quota ω_t^+ is equivalent to choosing an optimal shadow value λ^+, in the sense that the planner's first-order condition reduces to*

$$\lambda^+ = \beta \frac{1}{\sum_{j=1}^N y^j_{\lambda^+}} \mathbb{E}\Big[\sum_{i=1}^N (p^i - C^i(Z^i_t G^i(y^i_t))) Z^i_t G^i_{y^i_t} y^i_{\lambda^+}\Big] \tag{22}$$

which is identical in form to the tax condition (15)*. Both instruments target the same escapement vector $\bar{y}^i = \theta^i/(p^i - \tau^+)$.*

The equivalence is established in two ways in Appendix Appendix E. In the deterministic case, expression 20 becomes (22) because $\lambda_t = \beta \sum_i \rho^i_t V_{y^i}(\vec{y}_t)$. Thanks to duality, the manager can choose the target shadow value for convenience because it enjoys some properties of the tax basket:

Remark 4.1 (λ^+ is fixed over time) *Because $\lambda^+ = \tau^+$, then λ^+ is fixed over time.*

Remark 4.2 (Existence, uniqueness, interiority of the quota) *Through duality, λ^+ exists, is unique, and is interior under the same conditions of Corollary 2.3. Then, the quota ω_t exists, is unique, and is interior.*

The two instruments coincide at the planner's optimum but differ in the delivery. Under the tax the instrument fixes $\lambda = \tau^+$ directly, so $y^i(\tau^+)$ is deterministic. Under the quota, there is a difference between the shadow value of the fisherman λ^F_t, which is visible in the aggregate escapement definition:

$$\begin{aligned} \sum_i y^i_t &= \sum_i X^i_t - \omega_t \\ \sum_i Y^i(\lambda^F_t) &= \sum_i Z^i_t G^i(y^i_{t-1}) - \omega_t \end{aligned} \tag{23}$$

Remark 4.3 (The quota basket is not constant) *Under the tax, escapement $y^i_t = Y^i(\tau^+)$ is independent of the state variable. Under the quota, λ^F_t and y^i_t depend on the realized states through* (23)*. Because the manager lacks information on current stocks, the quota must be determined each period as y^i_{t-1} evolves, even though the optimal constant shadow value is known.*

Knowing λ^+ is convenient to determine the quota, but it doesn't capture the ecological shocks:

$$\omega^+_t = \sum_i X^i_t - \sum_i Y^i(\lambda^+), \tag{24}$$

Because the quota is set before harvest, the planner must estimate at least aggregate biomass. Reliable total stock information is expensive and contradictory, even when the basic information requirements to establish a species-specific regime are met. This weakens the presumption that quota baskets reduce monitoring requirements. Inferring aggregate conditions from an indicator species (as suggested by Karr et al., 2021) is not exempt from conducting its own stock assessment and introduces additional measurement error, further degrading the quota's performance under uncertainty.

Remark 4.4 (Information requirements) *Determining λ^+ (equivalently τ^+) requires no contemporaneous stock information. Implementing the tax requires none either. Implementing the quota $\omega_t = \sum_i G^i(y^i_{t-1}) - \sum_i Y^i(\lambda^+)$ requires at least learning the aggregate biomass, since the quota must be set before harvest.*

By the end of period t, comparing a complete-information and information-asymmetry quota basket shows the key difference between shadow values:

$$\sum_i (Z^i_t - 1)\, G^i(y^i_{t-1}) = \sum_i \Big[(C^i)^{-1}(p^i - \lambda^F_t) - (C^i)^{-1}(p^i - \lambda^+)\Big], \tag{25}$$

so λ^F_t inherits the randomness of the aggregate realized stock. It is possible to identify the source of the distortion:

Corollary 4.2 (The distortion between τ^+ and ω_t) *In a stochastic setting, the decomposition of the quota basket optimality condition*

$$\beta V_{y^k}(\vec{y}_{t-1}) = \pi^k(y^k_{t-1}) - \beta\,\mathrm{Cov}\big(R_t, Z^k_t\big)\, G^{k\prime}(y^k_{t-1}), \textit{where: } R_t = \lambda^F_t - \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t) \tag{26}$$

where the distortion is the covariance between the residual across shadow values and the ecological shocks for each i.

The full proof is available at Appendix E.5. The covariance vanishes in two cases: (i) the deterministic case; or (ii) marginal profits at the species-specific optima coincide. In every other case, the quota is suboptimal relative to the tax.

2.1.4 Welfare comparison

To compare the welfare outcomes across instruments, I borrow equation 17 (derived in Appendix C.1)

$$\mathcal{W} = \sum_i \underbrace{[A^i(X^i_0) - \kappa^i \frac{\beta}{1-\beta}]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \underbrace{[\phi^i(y^i_t) + \beta a^i(y^i_t)]}_{=:w^i(y^i_t)}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$ and $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$. Because Ξ_0 and the shock path are common to both instruments, I can cancel them in the difference. Then, I need to compare the per-species payoff w^i that I showed is strictly concave:

$$w^{i\prime}(y) = \pi^i(y) - \big(p^i - C^i(y)\big), \qquad w^{i\prime\prime}(y) = \phi^{i\prime\prime}(y) + \beta a^{i\prime\prime}(y) < 0 \tag{27}$$

So $w^{i\prime}(y^i) = 0$ when $\pi^i(\bar{y}^i) = p^i - C^i(\bar{y}^i) = \tau^+$, generating the vector of escapements that maximizes the intertemporal welfare. Under the tax, $y_t^i = Y^i(\tau^+) = \bar{y}^i$ is a fixed number, and the objective separates into a deterministic sequence of static problems for each i:

$$E[\mathcal{W}^\tau] = \Xi_0 + \frac{1}{1-\beta} \sum_i w^i(\bar{y}^i)$$

The quota does not separate. Escapement is random $y_t^i = Y^i(\lambda_t^F)$, function of every species' stock, with $\lambda_t^F = \lambda(\mathbf{X}_t, \omega_t^M)$ solving the extraction after the realization of the shocks Z_t^i,

$$y_t \;=\; \operatorname*{arg\,max}_{y \in \mathcal{H}_t \cap \mathcal{F}} \sum_i w^i(y^i), \qquad \mathcal{H}_t \;:=\; \Big\{ y \in \mathbb{R}_+^N \;:\; \textstyle\sum_i y^i = \sum_i Z_t^i G^i(\bar{y}^i) - \omega^M \Big\},$$

a random hyperplane. In contrast, the deterministic hyperplane $\mathcal{H}^* := \mathcal{H}_t\big|_{Z_t^i=1}$ corresponds to the tax basket's $\bar{y}$, since ω_t is set so that $E[\,\mathrm{RHS}_t\,] = \sum_i \bar{y}^i$ (through duality the planner targets $\mathcal{H}^*$).

Proposition 5 (The welfare is higher under a tax basket) *Let both instruments implement the same optimal shadow value $\lambda^+ = \tau^+$ (Corollary 4.1). Under the interior regularity condition and i.i.d. shocks with $E[Z_t^i] = 1$,*

$$E[\mathcal{W}^\tau] \;\geq\; E[\mathcal{W}^\omega]$$

The tax delivers $\sum_i w^i(\bar{y}^i)$ while the quota delivers $E\big[\sum_i w^i(Y^i(\lambda_t^F))\big]$. Since Ξ_0 cancels, summing the discounted per-period gaps,

$$E[\mathcal{W}^\tau] - E[\mathcal{W}^\omega] = \sum_{t \geq 0} \beta^t \, E\Big[\sum_i \big(w^i(\bar{y}^i) - w^i(y_t^i)\big)\Big]$$

Because w^i is strictly concave with global maximizer $\bar{y}^i$ (27), Jensen's inequality gives

$$E\Big[\sum_i w^i\big(Y^i(\lambda_t^F)\big)\Big] \;\leq\; \sum_i w^i\big(E[Y^i(\lambda_t^F)]\big) \;\leq\; \sum_i w^i(\bar{y}^i),$$

Equivalently, by Appendix E.5, the tax attains $\beta V^{\tau}_{y^k} = \pi^k(y^k_{t-1})$ exactly while the quota attains $\beta V^{\omega}_{y^k} = \pi^k(y^k_{t-1}) - \beta \operatorname{Cov}(R_t, Z^k_t)G^{k\prime}$. Because λ^F_t is a continuous random variable, $\Pr(\lambda^F_t = \lambda^+) = 0$, so $\Pr(y^i_t = \bar{y}^i) = 0$. Hence $\mathcal{H}_t \neq \mathcal{H}^*$ almost surely.

Under the assumptions of this paper, Weitzman's (2002) result holds: a tax outperforms a quota even under bundled management. However, even in this restricted context, the quota basket approaches the tax as the number of species grows. Remembering the discussion on the Corollary 2.3, I assume that baskets are balanced:

Remark 5.1 (Balanced basket) *The welfare convergence between instruments requires the basket to be balanced: no single species carries a non-vanishing share of the aggregate.*

Rename the $m\pi$ share to represent the tax basket case: $\alpha^i := \frac{(\bar{y}^i)^2/\theta^i}{\sum_j (\bar{y}^j)^2/\theta^j}$, with $\sum_i \alpha^i = 1$. In addition, define the fraction of an aggregate-quota shock absorbed by species i: $y^i_t - \bar{y}^i \approx \alpha^i S_t$. The basket is balanced if $\max_i \alpha^i \to 0$ as $N \to \infty$.

If one species dominates, with $\alpha^i \to \bar{\alpha} > 0$, its escapement variance $\operatorname{Var}(y^i_t) = (\alpha^i)^2 \operatorname{Var}(S_t) = \bar{\alpha}^2 \Theta(N)$ grows with N, its per-species loss stays $\Theta(1)$, and the total gap does not settle. Adding species helps only by spreading the aggregate shock across many comparably weighted stocks. A symmetric basket (all species are identical) has $\alpha^i = 1/N$. Taking this into account, I propose:

Proposition 6 (Convergence as $N \to \infty$) *Under a balanced basket (Remark 5.1 and the welfare-maximizing vector of $\bar{y}^i$, the per-species welfare difference between instruments decreases as $N \to \infty$.*

Comparing the "escapement budget" per instrument, $\sum_i y^i_t = RHS_t$ and $\sum_i \bar{y}^i = RHS^*$,

$$\sum_i (y^i_t - \bar{y}^i) = RHS_t - RHS^* = S_t, \qquad S_t = \sum_i (Z^i_t - 1) G^i(\bar{y}^i)$$

By variance additivity and independence,

$$\operatorname{Var}(S_t) = \sigma^2_Z \sum_i [G^i(\bar{y}^i)]^2 = \Theta(N),$$

so the species-specific variance becomes $\operatorname{Var}(S_t/N) = \Theta(1/N) \to 0$ by the law of large numbers.

For the quota basket, the shadow values converge, $E[(\lambda^F_t - \lambda^+)^2] = \Theta(1/N)$. Subtracting the realized and target budgets,

$$\sum_i \left[Y^i(\lambda^F_t) - Y^i(\lambda^+)\right] = S_t \tag{28}$$

Every Y^i is strictly increasing, so all N differences share the sign of $\lambda_t^F - \lambda^+$ and reinforce rather than cancel. Suppose $\lambda_t^F > \lambda^+$ (the reverse is symmetric). Then, by the mean value theorem, each difference is $Y^{i\prime}(\zeta^i)(\lambda_t^F - \lambda^+) \geq \underline{m}(\lambda_t^F - \lambda^+) > 0$ with the slope bound floor $\underline{m}$ of Lemma 4.2 (see Appendix F). After summing N deviations, I obtain:

$$S_t \geq \underline{m}\, N\, (\lambda_t^F - \lambda^+) \quad \Longrightarrow \quad |\lambda_t^F - \lambda^+| \leq \frac{|S_t|}{\underline{m}\, N}$$

Squaring and taking expectations,

$$E[(\lambda_t^F - \lambda^+)^2] \leq \frac{\mathrm{Var}(S_t)}{\underline{m}^2 N^2} = \frac{\Theta(N)}{\underline{m}^2 N^2} = \Theta(1/N) \to 0$$

Since $y_t^i = Y^i(\lambda_t^F)$ and $\bar{y}^i = Y^i(\lambda^+)$, the upper slope bound of Lemma 4.2 gives:

$$|y_t^i - \bar{y}^i| \leq \overline{m}^i\, |\lambda_t^F - \lambda^+| \quad \Longrightarrow \quad E[(y_t^i - \bar{y}^i)^2] \leq (\overline{m}^i)^2\, E[(\lambda_t^F - \lambda^+)^2] = O(1/N)$$

Because $\bar{y}^i$ is the basket maximum of w^i, Lemma 4.1 bounds the per-species loss by:

$$0 \leq w^i(\bar{y}^i) - w^i(y_t^i) \leq \tilde{B}^i\, (y_t^i - \bar{y}^i)^2, \qquad \tilde{B}^i := \tfrac{1}{2} \max_{[\underline{y}^i, \bar{y}^i]} \big(-w^{i\prime\prime}\big) < \infty$$

Taking expectations,

$$0 \leq w^i(\bar{y}^i) - E[w^i(y_t^i)] \leq \tilde{B}^i\, E[(y_t^i - \bar{y}^i)^2] = O(1/N)$$

The per-species loss is $O(1/N)$. Summing the N per-species losses,

$$0 \leq E[\mathcal{W}^\tau] - E[\mathcal{W}^\omega] = \sum_i \big(w^i(\bar{y}^i) - E[w^i(y_t^i)]\big) \leq \sum_i \tilde{B}^i\, E[(y_t^i - \bar{y}^i)^2] = \sum_i O(1/N) = O(1)$$

Adding species diversifies each shock at rate $1/N$.

3 Numerical results

To verify the theoretical comparison between the tax and quota baskets, I simulate welfare and biomass under each instrument using the same randomly generated fisheries and Monte-Carlo shock. I do not reproduce this analysis empirically to satisfy the regularity condition.

3.1 The dataset

Each replication draws a basket of N species. The objective is to satisfy the regularity condition under a cost function $C^i = \theta^i / s^i$:

Parameter	Meaning	Range / value
r^i	intrinsic growth rate	$U[0.30, 0.40]$
K^i	carrying capacity	[1200, 2400]
p^i	price	$\mathcal{U}[95, 105]$
$\kappa^i = e^{i*}/K^i$	escapement fraction	[0.48, 0.52]
θ^i	cost coefficient	from (30)
β	discount factor	$1/1.1 \approx 0.909$
σ_Z	shock volatility	0.20
T	horizon (periods)	40
N	basket size	$\{2, 10, 30, 50\}$
paths	Monte-Carlo paths	6000

Table 1: Parameters of the simulated fisheries. Initial stocks are set to the target escapement, $X_0^i = \bar{y}^i$, so both instruments start from the same state and make that all species are fishable.

$$\max_i \tau^{i*} \; < \; \tau^F_{\max} \; := \; \min_i \Big(p^i - \frac{\theta^i}{K^i} \Big), \tag{29}$$

Drawing naively $(r^i, K^i, p^i, \theta^i)$ becomes more challenging with N because the rejection rate explodes. This corroborates that species cannot be too heterogeneous (Corollary 2.3). To avoid this, I generate the data by fixing escapement as a fraction of its carrying capacity $\iota^i := y^{i*}/K^i$ and inverting Reed's for the cost coefficient:

$$\theta^i \; = \; \frac{p^i \left(\frac{1}{\beta} - G^{i\prime}(y^i) \right)}{\dfrac{1}{\beta y^i} - \dfrac{G^{i\prime}(y^i)}{G^i(e)}}. \tag{30}$$

By holding p^i and ι^i fixed, the tax does not vary with K^i. With this, I draw the variables and then adjust θ^i. The ranges for my variables are modest enough that no single species dominates the aggregate. Consequently, I keep prices and escapement-fraction ranges deliberately narrow. Growth and carrying capacity capture the biological heterogeneity. Growth shocks Z_t^i are i.i.d. lognormal with unit mean, $Z_t^i = \exp(\mu + \sigma_Z \varepsilon_t^i)$, $\varepsilon_t^i \sim \mathcal{N}(0, 1)$ and $\mu = -\sigma_Z^2/2$, so that $\mathbb{E}[Z_t^i] = 1$ and only the growth increment is perturbed, $x^i_{\text{pre}} = x^i + Z_t^i \, r^i x^i (1 - x^i/K^i)$.

3.2 Monte-Carlo considerations

For each N, I draw one basket and simulate both instruments over $T = 40$ periods on 6000 shock paths. Social welfare is the net present value of payoffs. For the tax, the social welfare also includes the net present tax revenue. I work with two variance controls: (i) common random numbers where tax and quota face identical shocks; and (ii) antithetic variates or mirror pairs such as ε and $-\varepsilon$. Together they reduce the standard error of the estimated welfare gap. My last check was verifying that the tax and quota yield identical welfare in a deterministic setting for heterogeneous baskets of size $N = 3, 17, 40$.

I work with three basket scenarios that take the welfare difference between a tax basket and a quota basket:

- Symmetric basket: every species is identical and, in consequence, $\tau^{i*} = \tau^{+}$.
- Balanced heterogeneous baskets: I draw r^i, K^i, p^i (and explained above, determine θ^i and $\bar{y}^i$) from the modest ranges (Table 1).
- Dominant basket: I force the share of one species in the aggregate biomass (60%) as N grows, so its idiosyncratic shock never diversifies away.

Finally, for the symmetric and balanced cases, I also compare the loss between the species-specific tax and each bundle instrument.

3.3 Results

Panel (a) in Figure 2 shows that the expected net present welfare difference $\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}]$ is positive at every N and converges to a constant: (i) around 209 welfare units in the symmetric basket and (ii) 220–230 in the balanced basket. In relative terms, the quota's loss is economically small (0.004% of $\approx$5.3 million) yet statistically sharp (over 100 standard errors from zero). The tax fixes $m\pi$ at the optimum τ^{+}, making each escapement constant, but not first-best every period. The quota fixes the quantity ex-ante and, when shocks happen, the shadow value adjusts to clear the realized biomass. λ_t^F is a random variable, and its dispersion around τ^{+} is the quota's second-best loss relative to the tax. This is the multispecies equivalent to the findings in Weitzman (2002). Panel (b) divides the result of (a) by N: the per-species difference vanishes at rate $O(1/N)$, following the c/N reference curve.

Figure 3(a) repeats the comparison of welfare with one species whose share of aggregate biomass is held fixed as N grows. If shocks never diversify, the per-species difference no longer vanishes (it stays near 125–240), and the total welfare difference grows with N (from about 250 at $N = 2$ to over 12,000 at $N = 50$). The loss is greater when diversification fails.

3.3.1 Decomposing the regulatory loss

The difference between instruments shows the second-best cost of using shared quantity control instead of price control. To complete the comparison, I use the species-specific first-best taxes (τ^i*), which are the stringent instruments targeting each individual $m\pi^i*$. The

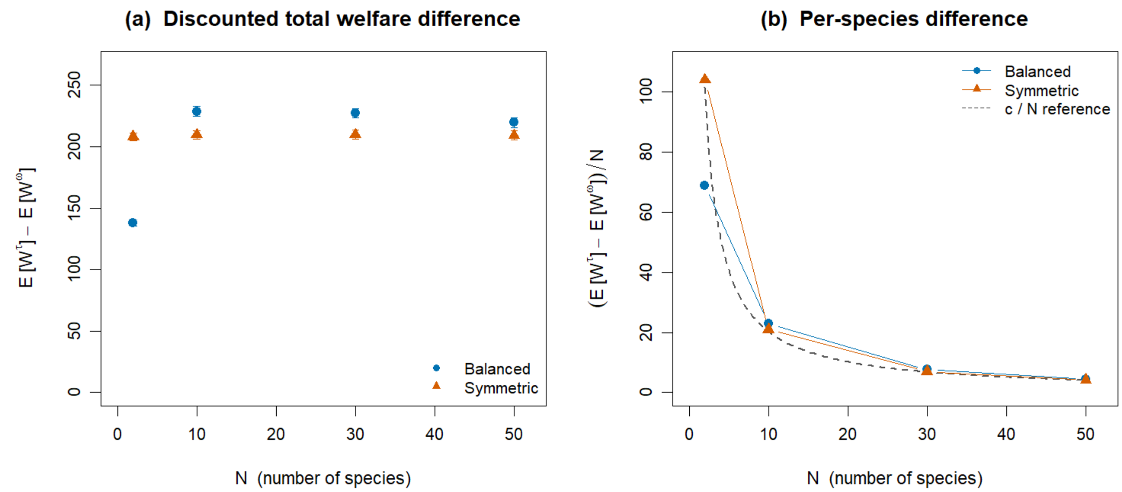

Figure 2: Left: expected total welfare difference $\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}]$ per basket size N, the difference converges to a positive constant. Right: the per-species difference vanishes at rate $O(1/N)$.

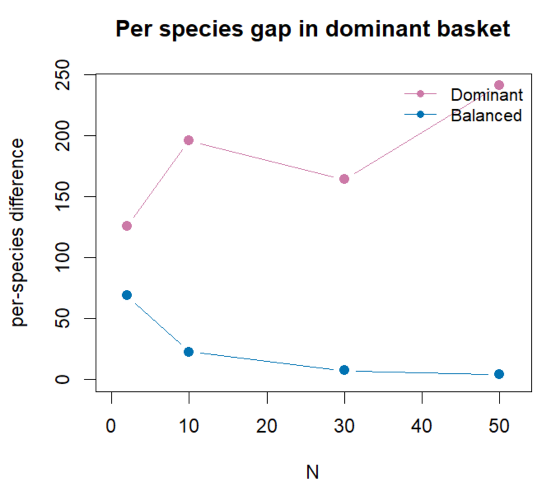

Figure 3: Robustness with one persistently dominant species. The per-species difference does not vanish, relative to the balanced basket.

sum of profits and tax revenues under this case generates the welfare W^{ind}. Because the first-best vector of taxes removes the losses associated with uncertainty and species pooling (forcing a shared instrument on heterogeneous individuals), I can order the welfare of each scheme as $W^{\text{ind}} \geq W^{\tau} \geq W^{\omega}$. With this consideration, I can decompose the regulatory loss

$$\underbrace{W^{\text{ind}} - W^{\omega}}_{\text{1. total loss}} = \underbrace{\left(W^{\text{ind}} - W^{\tau}\right)}_{\text{2. pooling loss}} + \underbrace{\left(W^{\tau} - W^{\omega}\right)}_{\text{3. uncertainty loss}} \tag{31}$$

These components go in opposite directions in N. The uncertainty loss can be spread with additional species, reaching a stable loss of 217 (25 averaged draws) dollars for all N, and a per-species fall of 110 dollars at $N = 2$ to about 4 dollars at $N = 50$. Meanwhile, the per-species loss in the pooling component increases from 1,400–2,000 dollars. This difference grows linearly in N: from 2,900 dollars at $N = 2$ to over 100,000 dollars at $N = 50$. Using bundle instruments creates an additional tradeoff between instrument mismatch costs and absorbing uncertainty. There will be a level of N where the pooling cost exceeds the uncertainty loss. This occurs at $N = 5$ for the current parameter ranges. This is progress relative to previous research that employed groups of 2 (Clark, 1990), 5 (Collado et al., 2021), or 6 (Collado et al., 2024) species. This suggests a possible interpretation of why the remaining biomass in open-access fisheries is stable in Chinese fisheries (Szuwalski et al., 2017). In practice, the Chinese fishing ban works like a tax basket because it doesn't allow for quota reallocation across species. However, it also implies that welfare losses from species pooling are high. This problem does not exist if all species are identical (symmetric basket), so similarity between traits appears to be a convenient and conservative recommendation (Collado et al., 2021, 2024; Kritzer et al., 2023).

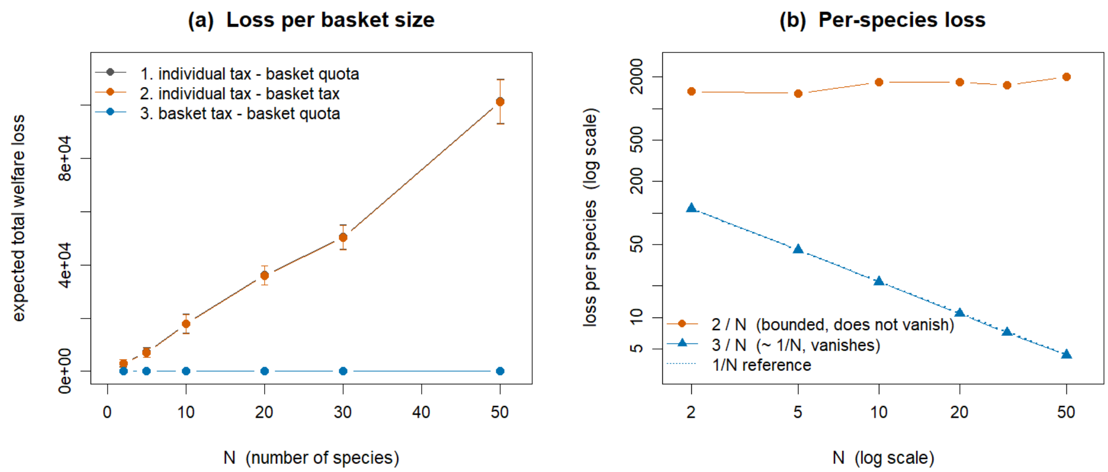

Figure 4: Decomposition of the total regulatory loss. Left: total losses; the 1. total and 2. pooling curves nearly coincide and grow with N, while the 3. uncertainty loss is flat near zero. Right: per-species losses on log–log axes: the 2. pooling loss per species is bounded away from zero while the 3. uncertainty loss per species vanishes at rate $O(1/N)$.

4 Discussion and conclusion

This paper studied the performance of basket instruments for managing heterogeneous fish stocks under ecological uncertainty. Its central contribution is showing one fundamental regulatory condition: equating marginal profits across all fishable species within the basket. In other words, catching one species alongside another naturally links the marginal value of the last unit caught. This condition integrates biological productivity, harvesting costs, and market prices. However, I also show that species cannot be too heterogeneous within the group because applying a coarse instrument comes at the cost of not applying species-specific regulation. This result helps to unify the outcomes in the recent literature on bundle instruments for fisheries because prior work has proposed designing quota baskets based on similarity of ecological traits (Costello and Szuwalski, 2018), biological or economic parameters (Collado et al., 2021), or relative vulnerability indices (Collado et al., 2024). This paper formalizes these insights by showing that a common instrument can regulate multiple heterogeneous fisheries, but at a trade-off between decreasing ecological uncertainty and losing aggregate performance (relative to a first-best policy) as the basket size increases.

The choice of instrument depends on how we target the aggregate shadow value $m\pi$. In general, both policies are second-best instruments, with tax baskets outperforming quota baskets in welfare. Both instruments are a weighted average of the marginal profits of each species, analogous to how uniform taxes are calculated for multi-pollutants (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). Because the problem is dynamic, the shares can change over time. In this context, a tax basket is particularly helpful because it fixes the remaining biomass (and shares) over time and is impervious to ecological shocks. This extends Weitzman's (2002) seminal work in landing fees versus quotas. While this result is attractive, the tax basket loses effectiveness relative to the species-specific instrument as N increases because the mismatch between the pooled and precise instruments grows. However, a tax basket must meet certain conditions to achieve first-best outcomes. The trivial one is that all species are identical or similar. The most relevant one is that each species marginal profits under its respective Reed's escapement are identical (in other words, each individual optimal tax is the same). The second case is also applicable to the quota basket, but in a deterministic setting.

Quota baskets, by contrast, inherit an additional distortion under uncertainty because the aggregate quota must be set before ecological shocks are realized. This timing asymmetry generally prevents quota baskets from outperforming the tax basket. However, when the basket includes many species and ecological shocks are independent, aggregation causes shocks to average out, reducing species-specific loss relative to the bundle tax. In this case, quota baskets can approximate the second-best tax. However, the quota basket also performs worse than the stringent instruments as basket size increases.

The trade-off between uncertainty diversification and policy mismatch motivates the policymaker to choose the right N. While this paper does not determine the optimal basket size, it offers recommendations. First, the planner must decide if the pooling loss is worth considering the system's administration costs. If basket size is large (as in trawling fisheries in China) and no monitoring capabilities exist, a bundle instrument would be recommendable as long as the total welfare loss does not exceed the regulatory expenses. On the other hand, if institutional and scientific capabilities are well-developed, a large basket is not a priority because the cost of monitoring one additional species can be relatively small (if the number of monitored species is already high). This would explain the stark differences between the basket experiences in California and China. On the other hand, when datasets are hard to build, species are hard to distinguish (as with the Canary and Yelloweye rockfish), or fishing gear is not selective enough, a basket instrument can be attractive. Keep in mind that this paper assumes all species are valuable and remain fishable. If species enter and exit the fishable set, a traditional management approach would be better.

On the specific information requirements: tax baskets depend only on structural parameters and optimal escapement targets and therefore do not require real-time stock-level information. It saves monitoring costs, but cost parameters are not easily available for all fisheries. Quota baskets, in contrast, require the planner to estimate current aggregate biomass to set the quota, weakening the common presumption that basket regulation necessarily reduces monitoring costs. While indicator species may partially alleviate this burden, they introduce additional measurement error that compounds ecological uncertainty and further degrades quota performance. [8]

These findings are particularly relevant for fisheries characterized by low selectivity, limited data, and high aggregation in landings. China's multispecies trawl fisheries illustrate this. These fisheries often harvest predators and prey jointly across trophic levels, with limited ability to identify or regulate individual stocks (Szuwalski et al., 2017). While such practices pose clear ecological risks, the results in this paper suggest that, in heavily aggregated systems with many exploited species, basket regulation may partially mitigate inefficiencies by smoothing ecological variability across stocks. This logic aligns with broader evidence that diversification across multiple stocks can reduce income volatility and targeting risk for fishers (Cline et al., 2017; Robinson et al., 2020). Conceptually, the result parallels findings in environmental economics showing that uniform instruments can perform well when uncertainty is diversified across regulated sources (Fowlie and Muller, 2019).

From a policy perspective, basket instruments are most attractive in data-limited settings where species-specific regulation is infeasible. However, they involve clear trade-offs. Quota baskets benefit from aggregation but require periodic updating and reliable information on aggregate biomass. Tax baskets, while more robust to uncertainty and informational constraints, achieve first-best outcomes only under restrictive conditions on marginal profits. These distinctions suggest that the choice between instruments should depend not only on ecological uncertainty but also on the regulator's informational capacity and monitoring costs.

Several limitations point to directions for future research. First, the analysis abstracts from technology choice and gear selectivity. Incorporating endogenous targeting decisions could clarify when basket instruments induce efficiency losses or gains through technological adjustment. Second, the framework focuses on a single basket; extending it to multiple, overlapping baskets raises questions about species mobility across regulatory groups and enforcement. Third, while species interactions are not modeled explicitly, preliminary derivations suggest the result is ambiguous. Fourth, the research doesn't consider bycatch of non-valuable or endangered species (e.g., Canary and Yelloweye rockfish). The latter case is particularly interesting because the model assumes a starting biomass that makes each species profitable. Establishing a bundle instrument could delay the recovery of the vulnerable species, but it would not heavily restrict extraction of the healthier stock.

Finally, linking the marginal-profit condition to trait-based or size-spectrum ecosystem models (Andersen et al., 2016) could help translate theoretical criteria into empirically observable metrics, informing practical basket design that balances efficiency, biodiversity, and risk.

In summary, tax baskets are generally more robust under uncertainty, while quota baskets can perform well in highly aggregated systems or under restrictive conditions. However,

[8] Sethi et al. (2005) demonstrated that measurement uncertainty has the greatest impact on fishery policy, profits, and fishery collapse under single-species management.

increasing basket size also increases instrument-mismatch loss. With careful attention to information requirements and ecological risk, basket regulation offers a pragmatic complement to species-specific management.

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Appendix A: Derivation of the optimal tax basket (1 species)

A.1: Fisherman's problem

The myopic fisherman maximizes profits through harvesting fish in period t:

$$\pi_t(x) = \max_{h_t \geq 0} \left\{ \int_{x_t - h_t}^{x_t} (p - \tau_t - C(s))\, ds \right\} \tag{32}$$

Where p is the unit price of fish, $C(s)$ represents the unit cost of harvesting (at fish population s), and τ is the landing fee per fish unit. After taking the first-order condition (FOC) concerning harvest, we obtain the optimality condition for the current period:

$$p - C\left(x_t - h_t\right) = \tau_t$$

Using the definition of escapement $y_t = x_t - h_t$, we can re-express this condition as:

$$p - C\left(y_t\right) = \tau_t \tag{33}$$

Notice that the fisherman takes an escapement decision:

$$\begin{aligned} p - \tau_t - C(y_t) &= 0 \\ C(y_t) &= p - \tau_t \\ y_t &= C^{-1}(p - \tau_t) \end{aligned} \tag{34}$$

A.2: Social planner's problem

The social planner tries to pick a landing fee to induce harvest levels that will maximize expected present-discounted profits, having the following form:

$$E[\sum_{t=0}^{\infty} \beta^t \int_{X_t - h_t}^{X_t} \pi(s) ds] \tag{35}$$

where $\beta \in \{0, 1\} \equiv \frac{1}{1+r}$ and the recruitment of fish stock during period t is denoted by $X_t = Z_t G(y_{t-1})$

A.3: Landing fee

The manager sets τ_t considering the fisherman decision function and the previous end-of-period escapement $C^{-1}(p - \tau_t)$ as the relevant state variable:

$$V(Y) = \max_{\tau \geq 0} E\left[\int_{C^{-1}(p - \tau_t)}^{Z_t G(C^{-1}(p - \tau_{t-1}))} \pi(s) ds + \beta V(Z_{t+1} G(C^{-1}(p - \tau_t))) \right] \tag{36}$$

Notice also that the profit reduction due to the tax and the tax revenue net each other. After this consideration, let's take the first order condition (FOC) concerning τ_t:

$$\frac{\partial V(.)}{\partial \tau_t} = 0 = E\left[-\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] + \beta E\left[\frac{d}{d\tau_t}[V_{t+1}(.)]\right] = 0 \tag{37}$$

Then, taking the FOC concerning τ_{t-1}, and using Leibniz integral rule:

$$\frac{\partial V_t(.)}{\partial \tau_{t-1}} = E\left[\left(p - C(Z_t G(y_{t-1}(\tau_{t-1})))\right)\frac{d}{d\tau_{t-1}}\left(Z_t G(y_{t-1}(\tau_{t-1}))\right)\right] \tag{38}$$

By Benveniste-Scheinkman theorem, I move forward the expression 7 by one period:

$$\frac{\partial V_{t+1}(.)}{\partial \tau_t} = E\left[\left(p - C(Z_{t+1} G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1} G(y_t(\tau_t))\right)\right]$$

Putting everything together:

$$\begin{aligned}
E\left[\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
E\left[\left(p - C(C^{-1}(p-\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \text{ Replace } y_t(\tau_t) \\
E\left[\left(p - (p-\tau_t)\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
E\left[(\tau_t)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[\frac{dZ_{t+1}}{d\tau_t}G(y_t(\tau_t)) + Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\right]\right]\frac{dy_t}{d\tau_t} \\
\tau_t &= \beta E\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t} - C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] \\
\tau_t &= \beta E\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] - \beta E\left[C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right] \\
\tau_t &= \beta p\frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1}C(Z_{t+1}G(y_t))\right]\frac{dG(y_t)}{dy_t}
\end{aligned}$$

In the derivation above, the expression is conditioned on period t information, and taxes do not influence shocks [9].

[9] A more stringent assumption is assuming independence between Z, G, and C

To make the result comparable with Reed (1979, I employ a cost function of the family $C(s) = \frac{\theta}{s^\alpha}$. Notice this function is homogeneous $C(ms) = \frac{\theta}{(ms)^\alpha} = m^{-\alpha}\frac{\theta}{(s)^\alpha}$. Then, the expression above can be expressed in two different ways according to catchability assumptions:

$$
\begin{aligned}
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1} C(Z_{t+1} G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1} Z_{t+1}^{-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1}^{1-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta \left[p - E\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t}
\end{aligned}
$$

For the element $E\left[Z_{t+1}^{1-\alpha}\right]$, I have 2 options on how to proceed:

- Assume constant catchability $\alpha = 1$, then $E\left[Z_{t+1}^{1-1}\right] = 1$, then the tax does not depend on the current stocks: $\tau_t = \beta\left[p - (C \circ G)(y_t)\right]\frac{dG(y_t)}{dy_t}$.
- Assume decreasing catchability $0 < \alpha < 1$, and then apply Jensen's Inequality: $E\left[Z_{t+1}^{1-\alpha}\right] \leq E\left[Z_{t+1}\right]^{1-\alpha} = 1$. Hence, $\tau_t \geq \beta\left[p - (C \circ G)(y_t)\right]\frac{dG(y_t)}{dy_t}$ is a lower bound (a minimum tax).

Notice that the LHS is equivalent to part of the Euler equation for constant escapement proposed by Clark (1976). Then, it is possible to propose a constant tax based on the constant optimal escapement that satisfies the Euler equation for y^*:

$$
\begin{aligned}
\tau_t &= \beta \left[p - E\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t} = (p - C(y_t)) \\
\tau^* &= \beta \left[p - E\left[Z_{t+1}^{1-\alpha}\right] C(G(y^*))\right] \frac{dG(y^*)}{dy^*} = (p - C(y^*)) \\
\tau^* &= \pi(y^*)
\end{aligned}
$$

The optimal tax depends only on the optimal escapement, as Weitzman (2002) shows.

Appendix B: Derivation of the optimal marginal profit $m\pi$

B.1: Fisherman's problem

In each period t, a myopic fisherman maximizes profits through harvesting N stocks $i = \{1, 2 \ldots, n\}$ with independent life cycles, considering that the planner uses an instrument that sets a target $m\pi_t$:

$$\begin{aligned} \pi_t = & \max_{\{h_t^i\}_{i=1}^N} \left\{ \sum_{i=1}^N \int_{X_t^i - h_t^i}^{X_t^i} \left(p^i - m\pi_t - C^i(s^i)\right) ds^i \right\} \\ & \text{s.t. } h_t^i \geq 0 \ \forall i \end{aligned} \tag{39}$$

where $m\pi_t$ is the target marginal profit that can be obtained through the landing fee per fish unit or a shared quota, applied to all species. From the FOC concerning each harvest, I obtain the optimality condition for the current period:

$$p^1 - C^1\left(X_t^1 - h_t^1\right) = p^2 - C^2\left(X_t^2 - h_t^2\right) = \cdots = p^n - C^n\left(X_t^n - h_t^n\right) = m\pi_t$$

Using the definition of escapement $y_t^i = X_t^1 - h_t^1$, I can re-express this condition as:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = m\pi_t \tag{40}$$

Notice that Equation 40 is the equimarginal principle: the total profits are maximized when the marginal profit of the last harvested unit for each species i is equal. Like in the previous section, the fisher sets the escapement decision:

$$\begin{aligned} p^i - m\pi_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - m\pi_t \\ y_t^i &= C^{i-1}(p^i - m\pi_t) \end{aligned} \tag{41}$$

The fisherman's escapement decision is independent of stock size and, therefore, unaffected by random ecological shocks.

B.2: Social planner's problem

The social planner incorporates the above fisherman's escapement decision into her dynamic optimization problem:

$$V_t(\vec{Y}_t) = \max_{\tau_t \geq 0} \mathbb{E}\left\{ \sum_{i=1}^N \int_{Y^i(m\pi_t)}^{Z_t^i G^i(Y_{t-1}^i(m\pi_{t-1}))} \left(p^i - C^i(s^i)\right) ds^i + \beta V_{t+1}(\vec{X}_{t+1})) \right\} \tag{42}$$

The target marginal profit depends on the available information. If the manager does not know the current stock status, I assume they operate based on each y_{t-1}^i. Although the planner cannot observe current ecological shocks, I include them in the dynamic problem to evaluate their impact on the tax basket. I take the FOC with respect to $m\pi_t$ (following the same procedure as in Appendix A) and obtain the following Euler equation:

$$\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y_t^i))y_{m\pi_t}^i\Big] = \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z_t^i G^i(y_t^i)))Z_t^i G_{y_t^i}^i y_{m\pi_t}^i\Big]$$

B.3: Tax basket equivalence

Replacing $m\pi = \tau_t$, it is straightforward to obtain the same expression as in Appendix B.2:

$$\tau_t = \beta\frac{1}{\sum_{i=1}^{N}\frac{dy_t^i}{d\tau_t}}\sum_{i=1}^{N}\left[p - E\left[Z_{t+1}^{i,1-\alpha}\right]C(G(y_t^i))\right]\frac{dG(y_t^i)}{dy_t^i}\frac{dy_t^i}{d\tau_t}$$

For the element $E\left[Z_{t+1}^{1-\alpha}\right]$, I have 2 options on how to proceed:

- Assume constant catchability $\alpha = 1$, then $E\left[Z_{t+1}^{1-1}\right] = 1$, then the tax does not depend on the current stocks.
- Assume decreasing catchability $0 < \alpha < 1$, and then apply Jensen's Inequality: $E\left[Z_{t+1}^{1-\alpha}\right] \leq E\left[Z_{t+1}\right]^{1-\alpha} = 1$. Hence, τ_t is a lower bound (a minimum tax).

The tax basket is the weighted average of the marginal profits per species in the next period. For simplicity, assuming $\alpha = 1$ and a cost function $C^i = \frac{\theta^i}{s^i}$ provides:

- $y_t^i = \frac{\theta^i}{p^i - \tau_t}$
- $\frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$

However, from the firm's optimization, I know that $p^i - \tau_t = C(y_t^i)$ and $C(y_t^i) = \frac{\theta^i}{y_t^i}$. With this information, I can rearrange the tax basket:

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p - E\left[Z_{t+1}^{i,1-\alpha} \right] C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t} \qquad \text{Replace } \alpha = 1$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{(p^i-\tau_t)^2}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{\theta^i}{(p^i - \tau_t)^2} \qquad \text{Replace } \frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{C^i(y_t^i)^2}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{\theta^i}{C^i(y_t^i)^2} \qquad \text{Replace } p^i - \tau_t = C^i(y_t^i)$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i)^2}{\theta^i}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{(y_t^i)^2}{\theta^i} \qquad \text{Replace } C^i(y_t^i) = \frac{\theta^i}{y_t^i}$$

$$\tau_t = \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i(\tau_t))^2}{\theta^i}} \sum_{i=1}^{N} \pi_{t+1}^i(y_t^i(\tau_t)) \frac{(y_t^i(\tau_t))^2}{\theta^i} \qquad \text{Rename next period's marginal profit}$$

$$\tau_t = \tau_t(\vec{y_t^i})$$

Similar to π_t, after employing Proposition 1 and Corollary 1.4, the tax basket is also a function of each species' escapement. If the species are identical, it is straightforward to verify that the tax basket equals the landing fee for individual species, as shown in Appendix A. Moreover, the social planner need not know the ecological shocks in the current period.

Appendix C: Auxiliary proofs

Appendix C.1: Welfare rearrangement

By Lemma 2.1 the fisherman equates $p^i - C^i(y_t^i) = \tau_t = \lambda_t$, so $y_t^i = m\pi_t = Y^i(\tau_t) = Y^i(\lambda_t)$. If $X_t^i > y_t^i$ for all i, this Lemma holds every period. Therefore, escapement depends on the current tax, not on the ecological shocks. With $C^i(s) = \theta^i/s$, the fisherman's myopic payoff is

$$\int_{y_t^i}^{X_t^i} \Big(p^i - \frac{\theta^i}{s}\Big) ds = A^i(X_t^i) + \phi^i(y_t^i), \quad \text{where } A^i(X) := p^i X - \theta^i \ln X, \quad \phi^i(y) := -p^i y + \theta^i \ln y,$$

which is 16. The current stock and the current escapement enter additively. Substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, using $\mathbb{E}[Z_t^i] = 1$, $\mathbb{E}|\ln Z_t^i| < \infty$. and $\ln(Z_t^i G^i) = \ln Z_t^i + \ln G^i$,

$$\begin{aligned}
\mathbb{E}\big[A^i\big(Z_t^i G^i(y_{t-1}^i)\big)\big] &= \mathbb{E}\big[p^i X - \theta^i \ln X\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln(Z_t^i G^i)\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln G^i - \theta^i \ln Z_t^i\big] \\
&= p^i G^i(y_{t-1}^i) - \theta^i \ln G^i(y_{t-1}^i) - \theta^i\, \mathbb{E}[\ln Z_t^i] \\
&= a^i(y_{t-1}^i) - \kappa^i
\end{aligned}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$ and $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$. Then, taking the initial stock X_0^i as given (and y_0^i as given) and summing across all species i,

$$\mathcal{W} = \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] \Big\}$$

Shifting the index in $\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i) = \beta \sum_{t\geq 0} \beta^t a^i(y_t^i)$ and reallocating β^t,

$$\begin{aligned}
\sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] &= \underbrace{\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i)}_{\text{(I): stock block}} - \underbrace{\sum_{t\geq 1} \beta^t \kappa^i}_{\text{(II): shock block}} + \underbrace{\sum_{t\geq 1} \beta^t \phi^i(y_t^i)}_{\text{(III): escapement block}} \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \sum_{t\geq 1} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \beta \sum_{t\geq 0} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i)
\end{aligned}$$

Putting things together:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \Big\} \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_i \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_i \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \sum_i \phi^i(y_t^i) \\
&= \sum_i [A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}] + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) + \sum_{t\geq 0} \beta^t \sum_i \phi^i(y_t^i) \\
&= \sum_i \underbrace{[A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i [\underbrace{\phi^i(y_t^i) + \beta a^i(y_t^i)}_{=:w^i(y_t^i)}]
\end{aligned}
$$

This outcome is applicable to a tax or quota basket.

Appendix C.2: Recovering the dynamic solution through the static problem

Reopening equation 17

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[\phi^i(y) + \beta a^i(y) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[\beta \big(p^i G^i(y) - \theta^i \ln G^i(y) \big) - \big(p^i y - \theta^i \ln y \big) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \underbrace{\frac{1}{1-\beta} \Big[\beta p^i G^i(y) - \beta \theta^i \ln G^i(y) - p^i y + \theta^i \ln y \Big]}_{\mathcal{V}^i} \Big\}
\end{aligned}
\tag{43}
$$

I take the derivative with respect to y:

$$
\begin{aligned}
\frac{d\mathcal{W}}{dy} &= \sum_i \frac{1}{1-\beta}\Big[\beta p^i G^{i\prime}(y) - \beta\theta^i \frac{1}{G^i(y)} G^{i\prime}(y) - p^i + \theta^i \frac{1}{y}\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\beta[p^i G^{i\prime}(y) - \theta^i \frac{1}{G^i(y)} G^{i\prime}(y)] - p^i + \theta^i \frac{1}{y}\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\beta[p^i - \theta^i \frac{1}{G^i(y)}] G^{i\prime}(y) - [p^i - \theta^i \frac{1}{y}]\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\beta[p^i - C^i(G^i(y))] G^{i\prime}(y) - [p^i - C^i(y)]\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\pi^i_{t+1}(y) - [p^i - C^i(y)]\Big]
\end{aligned}
\tag{44}
$$

I recover the Euler equation of Proposition 2. Moreover, ignoring the summation notation to focus on the species-specific management, the discount factor multiplies the growth block in 43; this is what makes 44 vanish exactly at Reed's y^{i*}. Because $\pi^{i\prime} \leq 0$ and $C^{i\prime} < 0$, so $\mathcal{V}^{i\prime\prime} < 0$: $\mathcal{V}^i$ is strictly concave with unique maximizer y^{i*}, then,

$$
\pi^i_{t+1}(y) > p^i - C^i(y) \;\text{ for } y < y^{i*}, \qquad \pi^i_{t+1}(y) < p^i - C^i(y) \;\text{ for } y > y^{i*} \tag{45}
$$

This component is key to proving uniqueness.

Appendix D: Existence, uniqueness and interiority of τ^+

By Proposition 2 and Corollary 2.2, the regularity condition $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$, and the cost function $C^i = \theta^i/s^i$, I can reinterpret the tax optimality condition as a residual function:

$$g(\tau) := \tau - R(\tau), \qquad \text{where: } R(\tau) := \frac{\sum_i \pi^i(Y^i(\tau))\,(Y^i(\tau))^2/\theta^i}{\sum_i (Y^i(\tau))^2/\theta^i},$$

by Equation 18, $g(\tau)$ and $\mathcal{W}'(\tau)$ share sign and roots.

Appendix D.1: Endpoint signs

From Corollary 2.2 (Appendix C.2), I know the tax solution commits species i to a constant escapement y forever. Without loss of generality, order species according to their individual optimal taxes $\tau^{1*} \leq \cdots \leq \tau^{N*}$ and assume not all τ^{i*} are equal (the equal case is Remark **??**). Then, I can establish the following logical sequence:

1. $Y^i(\cdot)$ is strictly increasing, so $\tau < \tau^{i*} \iff Y^i(\tau) < y^{i*}$ (lower tax, less remaining stock in the ocean).
2. At $\tau = \tau^{1*}$: species 1 contributes exactly $\pi^1_{t+1}(Y^1(\tau^{1*})) = \tau^{1*}$ (shown in Appendix A).
3. Each $j \geq 2$ has $\tau^{1*} < \tau^{j*}$, hence $Y^j(\tau^{1*}) < y^{j*}$ (over-harvest), and 45 with $p^j - C^j(Y^j(\tau^{1*})) = \tau^{1*}$ gives $\pi^j_{t+1}(Y^j(\tau^{1*})) > \tau^{1*}$.
4. A weighted average with strictly positive weights (thanks to the fishable range) that are all $\geq \tau^{1*}$, at least one strictly, exceeds τ^{1*}. Therefore $R(\tau^{1*}) > \tau^{1*}$ and $g(\tau^{1*}) < 0$.
5. Starting from the opposite endpoing, $\tau = \tau^{N*}$: symmetrically, each $j < N$ has $Y^j(\tau^{N*}) > y^{j*}$ (under-harvest) so $\pi^j_{t+1}(Y^j(\tau^{N*})) < \tau^{N*}$, giving $R(\tau^{N*}) < \tau^{N*}$ and $g(\tau^{N*}) > 0$.

Appendix D.2: Existence

Y^i, π^i and the denominator $\sum_i \theta^i/(p^i - \tau)^2$ are continuous on $[\tau^{1*}, \tau^{N*}]$. The interior regularity guarantees $p^i - \tau \geq C^i(K^i) > 0$; then the denominator is bounded and bounded away from zero, and g is continuous. By the Intermediate Value Theorem there exists $\tau^+ \in (\tau^{1*}, \tau^{N*})$ with $g(\tau^+) = 0$.

Appendix D.3: Uniqueness

The sufficient condition is that $\mathcal{W}$ be strictly concave on $[\tau^{1*}, \tau^{N*}]$ as shown in Appendix C.2. However, concavity of each w^i in its respective escapement is not enough, because τ enters w^i through Y^i and $Y^{i\prime}$ sets the weights. Without loss of generality, I define and focus on $W(\tau) = \sum_i w^i(Y^i(\tau))$. Differentiating W twice:

$$W''(\tau) = \sum_i \Big[\underbrace{w^{i\prime\prime}(y^i)\left(Y^{i\prime}(\tau)\right)^2}_{\leq 0 \text{ if } w^i \text{ concave}} + \underbrace{w^{i\prime}(y^i)\, Y^{i\prime\prime}(\tau)}_{\text{indefinite}}\Big], \qquad y^i = Y^i(\tau) \tag{46}$$

Notice that y^i differ across species even though τ is shared, and the sum contains no cross term (W is separable). Then, the global concavity is equivalent to per-species concavity. However, the term $w^{i\prime}(y^i)\, Y^{i\prime\prime}(\tau)$ has ambiguous sign: $Y^{i\prime\prime}(\tau) = 2\theta^i/(p^i - \tau)^3 > 0$, while $w^{i\prime}(y^i) = \pi^i_{t+1}(y^i) - \tau$ is positive below τ^{i*} and negative above it by 45.

To treat this, I remove the nonlinearity of Y^i through affine reparametrization. From the fisherman's rule $p^i - \theta^i/y^i = \tau$,

$$\begin{aligned} p^i - \frac{\theta^i}{y^i} &= \tau \\ z^i := \frac{1}{y^i} &= \frac{p^i - \tau}{\theta^i} \end{aligned}$$

is affine in τ, so $z^{i\prime\prime}(\tau) = 0$ and the curvature term vanishes. With this, I rewrite W:

$$W(\tau) = \sum_i w^i(y^i) = \sum_i w^i(1/z^i) = \sum_i \hat{w}^i(z^i), \quad \text{where } \hat{w}^i(z) := w^i(1/z).$$

Because each z^i is affine in τ and W is separable, $W''(\tau) = \sum_i \hat{w}^{i\prime\prime}(z^i)\,(z^{i\prime})^2$ with $(z^{i\prime})^2 > 0$. Then, W is strictly concave in $\tau \iff \hat{w}^{i\prime\prime}(z^i) \leq 0$ for all i. Using $\dfrac{dy}{dz} = \dfrac{d}{dz} z^{-1} = -\dfrac{1}{z^2}$ and $z = 1/y \Rightarrow \dfrac{1}{z^2} = y^2$, I get,

$$\hat{w}^{i\prime}(z) = w^{i\prime}\left(\tfrac{1}{z}\right)\left(-\tfrac{1}{z^2}\right) = -\tfrac{1}{z^2}\, w^{i\prime}(y)$$

and

$$\hat{w}^{i\prime\prime}(z) = \tfrac{2}{z^3} w^{i\prime}\left(\tfrac{1}{z}\right) - \tfrac{1}{z^2} w^{i\prime\prime}\left(\tfrac{1}{z}\right)\left(-\tfrac{1}{z^2}\right) = \tfrac{2}{z^3} w^{i\prime}\left(\tfrac{1}{z}\right) + \tfrac{1}{z^4} w^{i\prime\prime}\left(\tfrac{1}{z}\right).$$

Replacing $z = 1/y$ (so $1/z^3 = y^3$, $1/z^4 = y^4$),

$$\begin{aligned} \hat{w}^{i\prime\prime}(z) &= 2y^3 w^{i\prime}(y) + y^4 w^{i\prime\prime}(y) \\ &= y^3\left[\, y\, w^{i\prime\prime}(y) + 2w^{i\prime}(y)\,\right] \end{aligned}$$

From Corollary 2.2, I know $w^{i\prime}(y) = \pi^i_{t+1}(y) - p^i + \theta^i/y$ and $w^{i\prime\prime}(y) = \pi^{i\prime}_{t+1}(y) - \theta^i/y^2$,

$$\begin{aligned}
\hat{w}^{i\prime\prime}(z) &= y^3\left[\, y\,(\pi^{i\prime}_{t+1}(y) - \theta^i/y^2) + 2(\pi^i_{t+1}(y) - p^i + \theta^i/y)\right] \\
&= y^3\left[y\pi^{i\prime}_{t+1}(y) - \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i + 2\theta^i/y\right] \\
&= y^3\left[y\pi^{i\prime}_{t+1}(y) + \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i\right] \\
&= y^3\left[y\pi^{i\prime}_{t+1}(y) + C^i(y) + 2\pi^i_{t+1}(y) - 2p^i\right].
\end{aligned}$$

Using the fisherman's species-specific optimality condition $C^i(Y^i(\tau)) = p^i - \tau$,

$$\begin{aligned}
\hat{w}^{i\prime\prime}(z) &= y^3\left[y\pi^{i\prime}_{t+1}(y) + p^i - \tau + 2\pi^i_{t+1}(y) - 2p^i\right] \\
&= \underbrace{y^3}_{>0}\, \underbrace{\left[y\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) - p^i - \tau\right]}_{\text{sign of } \hat{w}^{i\prime\prime}}.
\end{aligned}$$

Since $y^3 > 0$, $\hat{w}^{i\prime\prime}(z) \leq 0$ if and only if the bracket is ≤ 0, which is:

Condition for global concavity For every i and every $\tau \in [\tau^{1*}, \tau^{N*}]$, writing $y = Y^i(\tau)$,

$$y\,\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) \;\leq\; p^i + \tau, \tag{47}$$

with strict inequality for at least one i. Because W is separable, imposing (47) for every i makes $W''(\tau) \leq 0$.

Appendix D.4: Interiority

The endpoint signs $g(\tau^{1*}) < 0 < g(\tau^{N*})$ are strict, so $\tau^+ \in (\tau^{1*}, \tau^{N*}) \subset (0, \tau^F_{\max})$. Then, the optimum is never at $\min_i \tau^{i*}$ or $\max_i \tau^{i*}$ unless these coincide.

Appendix E: Derivation of the optimal quota basket

E.1: Fisherman's problem

Each period, the fisherman maximizes profits over the harvest of N stocks subject to the shared quota:

$$\begin{aligned} &\max_{h^i} \left\{ \pi_t = \sum_i p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s_t^i)\, ds^i \right\} \\ &\text{subject to} \quad \omega_t = \sum_i h_t^i = \sum^2 X_t^i - \sum^2 y_t^i \end{aligned} \tag{48}$$

The Lagrangian is:

$$\max_{h^i} \left\{ \sum_{i=1} p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s_t^i)\, ds^i - \lambda_t \Big(\omega_t - \sum h_t^i \Big) \right\},$$

with FOC for each stock

$$\begin{aligned} p^i - C^i(y_t^i) &= \lambda_t \\ y_t^i &= (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t) \end{aligned} \tag{49}$$

Due to assumption vii, the realized stocks $X_t^i = Z_t^i G^i(y_{t-1}^i)$ depend on last period's escapement and this period's shock, not on the current quota:

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\ \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \lambda_t &= \lambda(\mathbf{X}_t, \omega_t) \end{aligned} \tag{50}$$

Differentiating (50) with respect to ω_t, holding the predetermined stocks $\mathbf{X}_t$ fixed (because the fisherman observes them),

$$\begin{aligned} \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \frac{\partial}{\partial \omega} \Big(\sum_i Y^i(\lambda_t) \Big) &= \frac{\partial}{\partial \omega} \Big(\sum_i X_t^i - \omega_t \Big) \\ \Big(\sum_i Y_\lambda^i \Big) \lambda_{\omega_t} &= -1 \\ \lambda_{\omega_t} &= \frac{-1}{\sum_i Y_\lambda^i} < 0 \end{aligned} \tag{51}$$

If the cost function is $C^{=}\theta^i/s^i$, $Y^i_\lambda = \theta^i/(p^i-\lambda)^2 = (y^i)^2/\theta^i > 0$ because $p^i - \lambda = C^i(y^i) = \theta^i/y^i$ and $y^i, \theta^i > 0$ by definition. Consequently, a larger quota lowers the common shadow value. A change in the harvest limit causes the following change in the species-specific escapement:

$$
\begin{aligned}
y^i_t &= Y^i(\lambda_t) \\
\frac{\partial y^i_t}{\partial \omega_t} &= Y^i_\lambda \, \lambda_{\omega_t} \\
&= Y^i_\lambda \; \frac{-1}{\sum_j Y^j_\lambda} && \text{Replace (51)} \\
&= -\frac{(y^i_t)^2/\theta^i}{\sum_j (y^j_t)^2/\theta^i} \\
&= -\rho^i_t && \text{Rename}
\end{aligned}
\tag{52}
$$

Notice that $\sum_i \rho^i_t = 1$ if the regularity condition holds. The weights ρ^i_t record how the fisherman spreads one extra unit of quota basket across species.

E.2: Choosing the quota is choosing the shared shadow value

The total profits per period are:

$$
\Pi(\mathbf{X}_t, \omega_t) := \sum_i \int_{y^i_t}^{X^i_t} \left(p^i - C^i(s)\right) ds
$$

with $y^i_t = Y^i(\lambda_t)$. Deriving with respect to ω, using the fisherman's FOC $p^i - C^i(y^i_t) = \lambda_t$ and $\sum_i \partial y^i_t/\partial \omega_t = -1$ from (50),

$$
\begin{aligned}
\frac{\partial \Pi}{\partial \omega_t} &= -\sum_i \left(p^i - C^i(y^i_t)\right) \frac{\partial y^i_t}{\partial \omega_t} \\
&= -\sum_i \lambda_t \frac{\partial y^i_t}{\partial \omega_t} \\
&= -\lambda_t \sum_i \frac{\partial y^i_t}{\partial \omega_t} \\
&= -\lambda_t \sum_i (-\rho^i_t) \\
&= \lambda_t
\end{aligned}
\tag{53}
$$

The marginal social value of an extra unit of aggregate quota equals the fisherman's shadow value λ_t. This is the duality between the two derivations: λ_t is the price that decentralizes the quota.

E.3: Social Planner's problem

Take the predetermined escapement vector $\vec{y}_{t-1}$ as the state. The planner sets the quota ω_t before the shocks, so ω_t is deterministic while λ_t and y_t^i are random through $\mathbf{X}_t$:

$$V(\vec{y}_{t-1}) = \max_{\omega_t \geq 0} \mathbb{E}\Big[\Pi(\mathbf{X}_t, \omega_t) + \beta V(\vec{y}_t)\Big], \qquad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad y_t^i = Y^i\big(\lambda(\mathbf{X}_t, \omega_t)\big) \tag{54}$$

Only y_t^i depends on ω_t (the stocks $\mathbf{X}_t$ are predetermined on y_{t-1}^i). Using $\partial\Pi/\partial\omega_t = \lambda_t$ from 53 and $\partial y_t^i/\partial\omega_t = -\rho_t^i$ from 52, I derive the FOC concerning the quota basket,

$$\begin{aligned} \frac{\partial V(\cdot)}{\partial \omega_t} &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta[V_{y^1} y_\lambda^1 \lambda_\omega + V_{y^2} y_\lambda^2 \lambda_\omega + ... + V_{y^n} y_\lambda^n \lambda_\omega]\Big] = 0 \\ &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta \sum_j V_{y^j}(\vec{y}_t) \frac{\partial y_t^j}{\partial \omega_t}\Big] = 0 \\ &= \mathbb{E}\Big[\lambda_t - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big] = 0 \end{aligned} \tag{55}$$

The optimal quota basket equates, in expectation, the current shadow value of the quota with the ρ-weighted continuation value of the escapement.

The state y_{t-1}^k enters 54 through $X_t^k = Z_t^k G^k(y_{t-1}^k)$. Differentiating 50 with respect to the stock, holding ω_t fixed,

$$\Big(\sum_j Y_\lambda^j\Big) \frac{\partial \lambda_t}{\partial X_t^k} = 1, \qquad \frac{\partial y_t^j}{\partial X_t^k} = Y_\lambda^j \frac{\partial \lambda_t}{\partial X_t^k} = \frac{Y_\lambda^j}{\sum_m Y_\lambda^m} = \rho_t^j, \qquad \sum_j \frac{\partial y_t^j}{\partial X_t^k} = 1 \tag{56}$$

Applying the envelope theorem (the indirect effect through the optimal ω_t is zero by 55) and the chain rule,

$$
\begin{aligned}
V_{y^k_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(\frac{\partial \Pi}{\partial X^k_t} + \beta \frac{\partial V(\vec{y}_t)}{\partial X^k_t}\Big) \underbrace{Z^k_t G^{k\prime}(y^k_{t-1})}_{\partial X^k_t/\partial y^k_{t-1}}\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\frac{\partial X^i_t}{\partial X^k_t} - \sum_i \big(p^i - C^i(y^j_t)\big)\frac{\partial y^j_t}{\partial \lambda_t}\frac{\partial \lambda_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^i_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\delta_{ik} - \sum_i \big(p^i - C^i(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^i_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)1 - \sum_i \big(p^i - C^i(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^i_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t \sum_j \rho^j_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big]
\end{aligned}
\tag{57}
$$

using $\partial X^i_t/\partial X^k_t = \delta_{ik}$ for the stocks (only stock k responds to its respective y^k_{t-1}) and $\partial y^i_t/\partial X^k_t = \rho^i_t$ from 56 for the quota-coupled escapements (all respond, through λ_t). The bracket is the shadow value of one extra fish in stock k minus the λ_t of the escapement it displaces, plus its continuation value. The escapements are all tied through the quota basket.

Equations 55–57 are the direct-ω analog to Appendix 4.

E.4: Recovering λ^+

With $Z^i_t \equiv 1$ the expectation drops and 55 holds pointwise, $\lambda_t = \beta \sum_i \rho^i_t V_{y^i}(\vec{y}_t)$. Substituting this into 57 cancels the last two bracketed terms, leaving

$$
\begin{aligned}
V_{y^i_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}_{Z_t}\Big[[p^i - C^i(X^i_t) - \lambda_t + \beta \sum_i \rho^i_t V_{y^i}(\vec{y}_t)] Z^i_t G^{i\prime}(y^i_{t-1})\Big] \\
&= \Big[[p^i - C^i(X^i_t) - \lambda_t + \lambda_t] G^{i\prime}(y^i_{t-1})\Big] \\
&= [p^i - C^i(X^i_t)] G^{i\prime}(y^i_{t-1}) \\
&= \frac{1}{\beta}\pi^i(y^i_{t-1}) \quad \text{Using definition of } \pi^i = [p^i - C^i]G^{i\prime}
\end{aligned}
\tag{58}
$$

Substituting back into the FOC and using the weights 1.4,

$$
\begin{aligned}
\lambda_t &= \beta \sum_i \rho_t^i V_{y^i}(\vec{y}_t) \\
\lambda_t &= \sum_i \rho_t^i \, \pi^i(y_t^i) \\
\lambda^+ &= \frac{\sum_i \pi^i\big(y_t^i\big)\,(y_t^i)^2/\theta^i}{\sum_j (y_t^j)^2/\theta^j}
\end{aligned}
\tag{59}
$$

which is exactly the planner's FOC for the tax 15. Like the optimal tax basket, the optimal shadow value is fixed over time $\lambda_t = \lambda^+$ with $y_t^i = Y^i(\lambda^+)$ if the regularity condition holds.

E.5: Quota basket is a second best

Under uncertainty, the collapse of Appendix E.4 fails because the shadow value λ_t^F and the reallocation weights ρ_t^i are both functions of the realized shocks. The planner, by contrast, commits to a deterministic ω_t. The FOC residual is:

$$
R_t := \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t),
$$

so that 55 is $\mathbb{E}[R_t] = 0$. Substituting into 57 and separating,

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - R_t\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\big[(p^k - C^k(X_t^k))Z_t^k\big] G^{k\prime}(y_{t-1}^k) - \mathbb{E}\big[R_t Z_t^k\big] G^{k\prime}(y_{t-1}^k)
\end{aligned}
$$

With $C^k(s) = \theta^k/s$ and $X_t^k = Z_t^k G^k(y_{t-1}^k)$ the shock cancels in the first expectation,

$$
\mathbb{E}\Big[\Big(p^k - \frac{\theta^k}{Z_t^k G^k(y_{t-1}^k)}\Big) Z_t^k\Big] = p^k - \frac{\theta^k}{G^k(y_{t-1}^k)} = p^k - C^k\big(G^k(y_{t-1}^k)\big)
$$

However, the second expression is a covariance

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \frac{1}{\beta}\pi_{t+1}^k(y_{t-1}^k) - \mathrm{Cov}\big(R_t,\, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k) \\
\beta V_{y^k}(\vec{y}_{t-1}) &= \pi_{t+1}^k(y_{t-1}^k) - \beta \mathrm{Cov}\big(R_t,\, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k)
\end{aligned}
\tag{60}
$$

In a deterministic setting, $\mathrm{Cov}(R_t, Z_t^k) = 0$ and 60 reduces to $\beta V_{y^k} = \pi^k$, which is Appendix E.4. Under uncertainty, the covariance is nonzero, because R_t moves with the shocks through both λ_t^F and ρ_t^j. The planner's single scalar ω_t can set the mean of R_t

to zero, but cannot remove its correlation with the shock vector. Given the deterministic quota,

$$\sum_i Y^i(\lambda_t^F) = \sum_i Z_t^i G^i(y_{t-1}^i) - \omega_t$$

so λ_t^F inherits the randomness of the aggregate realized stock. The quota choice

$$\omega_t = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\lambda^+) \tag{61}$$

uses only $\mathbb{E}[Z_t^i] = 1$ and the predetermined y_{t-1}^i. It centers the distribution of λ_t^F near λ^+ and the escapements near their targets $\bar{y}^i = Y^i(\lambda^+)$, and coincides with the ω_t.

Appendix F: the bounds of the welfare loss

Lemma 4.1 (The welfare loss is bounded) *$w^i(y) = \phi^i(y) + \beta a^i(y)$ is the dynamic per-species payoff of (17), with $\phi^i(y) = -p^i y + \theta^i \ln y$ and $a^i(y) = p^i G^i(y) - \theta^i \ln G^i(y)$. By (27), $w^{i\prime}(y) = \pi^i_{t+1}(y) - (p^i - C^i(y))$, so $w^{i\prime}(\bar{y}^i) = 0$, and $w^{i\prime\prime}(y) < 0$ on the fishable range $[\underline{y}^i, \hat{y}^i]$, where the bounds are the smallest and largest escapements the quota can induce (see Lemma 4.2), so that $0 < \underline{y}^i \le \bar{y}^i \le \hat{y}^i < \infty$. Then, for every $y \in [\underline{y}^i, \hat{y}^i]$:*

(a) The loss relative to the basket optimum is non-negative:

$$0 \;\le\; w^i(\bar{y}^i) - w^i(y). \tag{62}$$

(b) The loss is at most a constant times the squared deviation:

$$w^i(\bar{y}^i) - w^i(y) \;\le\; \tilde{B}^i\,(y - \bar{y}^i)^2, \qquad \tilde{B}^i = \tfrac{1}{2}\max_{[\underline{y}^i, \hat{y}^i]}\big(-w^{i\prime\prime}\big) < \infty. \tag{63}$$

The global maximizer of the concave $\bar{y}^i$ verifies (a). For part (b), I know:

$$w^i(b) - w^i(y) = \int_y^b w^{i\prime}(s)ds$$

renaming $u = w^{i\prime}(s)$ and $v = s - b$, then by integration by parts:

$$\begin{aligned}
\int_y^b w^{i\prime}(s)ds &= \Big[w^{i\prime}(s)(s-b)\Big]\Big|_b^y - \int_y^b (s-b)w^{i\prime\prime}(s)ds \\
&= -w^{i\prime}(y)(y-b) - \int_y^b (s-b)w^{i\prime\prime}(s)ds \\
&= w^{i\prime}(y)(b-y) - \int_y^b (s-b)w^{i\prime\prime}(s)ds
\end{aligned}$$

if $b = \bar{y}^i$, then $w^{i\prime} = 0$, then

$$w^i(\bar{y}^i) - w^i(y) = \int_y^{\bar{y}^i} \big(-w^{i\prime\prime}(s)\big)\,(s-y)\,ds,$$

The integrand is non-negative because $w^{i\prime\prime} < 0$. I control the curvature of $w^{i\prime\prime}$ with the maximum possible value in the range:

$$M^i := \max_{[\underline{y}^i, \hat{y}^i]}(-w^{i\prime\prime})$$

Then, for every s, $-w^{i\prime\prime}(s) \le M^i$. Since $(s - y) \ge 0$, then

$$-w^{i\prime\prime}(s)(s-y) \leq M^i(s-y)$$

Integrating both sides

$$\begin{aligned}
-w^{i\prime\prime}(s)(s-y) &\leq M^i(s-y) \\
\int_y^{\bar{y}^i} -w^{i\prime\prime}(s)(s-y)ds &\leq M^i \int_y^{\bar{y}^i} (s-y)ds \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \int_y^{\bar{y}^i} (s-y)ds \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \int_y^{\bar{y}^i - y} udu \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \frac{u^2}{2}\Big|_{\bar{y}^i - y}^{0} \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \frac{(\bar{y}^i - y)^2}{2} \\
0 \leq w^i(\bar{y}^i) - w^i(y) &\leq \tilde{B}^i(y - \bar{y}^i)^2
\end{aligned}$$

The maximum is finite because $w^{i\prime\prime}$ is continuous and no species can be driven to extinction due to the regularity condition.

The next lemma shows the shadow-value deviation is also bounded:

Lemma 4.2 (Bounds for the fisherman's response) *Let $Y^i(\lambda) = \theta^i/(p^i - \lambda)$ be the fisherman's escapement response under a quota. Suppose the realized shadow value is confined to a bounded interval $\lambda_t^F \in [\lambda_L, \lambda_u]$ with $\lambda_u < \min_i p^i$. Then there exist finite positive constants*

$$\underline{m} := \min_i \frac{\theta^i}{(p^i - \lambda_L)^2} > 0, \qquad \overline{m}^i := \frac{\theta^i}{(p^i - \lambda_u)^2} < \infty, \tag{64}$$

such that, for any two shadow values $\lambda, \lambda' \in [\lambda_L, \lambda_u]$,

$$\underline{m}\,|\lambda - \lambda'| \;\leq\; \big|Y^i(\lambda) - Y^i(\lambda')\big| \;\leq\; \overline{m}^i\,|\lambda - \lambda'|. \tag{65}$$

In particular, taking $\lambda = \lambda_t^F$, $\lambda' = \lambda^+$ and $y_t^i = Y^i(\lambda_t^F)$, $\bar{y}^i = Y^i(\lambda^+)$,

$$\underline{m}\,|\lambda_t^F - \lambda^+| \;\leq\; |y_t^i - \bar{y}^i| \;\leq\; \overline{m}^i\,|\lambda_t^F - \lambda^+|. \tag{66}$$

The response function has derivative

$$Y^{i\prime}(\lambda) = \frac{d}{d\lambda}\frac{\theta^i}{p^i - \lambda} = \frac{\theta^i}{(p^i - \lambda)^2} > 0,$$

so Y^i is strictly increasing. Differentiating once more, $Y^{i\prime\prime}(\lambda) = 2\theta^i/(p^i - \lambda)^3 > 0$ on $\lambda < p^i$, so $Y^{i\prime}$ is itself increasing in λ. Then, the response is flattest at the lower end λ_L and steepest at the upper end λ_u. Hence on $[\lambda_L, \lambda_u]$,

$$\frac{\theta^i}{(p^i - \lambda_L)^2} \;\leq\; Y^{i\prime}(\lambda) \;\leq\; \frac{\theta^i}{(p^i - \lambda_u)^2} = \overline{m}^i$$

Taking the minimum over species gives the floor $\underline{m}$ in (64). Both constants are finite and strictly positive because $\lambda_u < \min_i p^i$ avoids the denominator becoming 0.

By the mean value theorem, for any λ, λ' there is a point ζ^i between them with $Y^i(\lambda) - Y^i(\lambda') = Y^{i\prime}(\zeta^i)(\lambda - \lambda')$. Since $\zeta^i \in [\lambda_L, \lambda_u]$, its slope lies between the two bounds, which gives (65).